Student's Guide in

Mathematics

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First-year Secondary

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Contents

Unit 3: Real function

Lesson 1: Real function.
Lesson 2: Domain and range of a function.
Lesson 3: Constant function and linear function.
Lesson 4: Quadratic function.
Lesson 5: Cubic function.
Lesson 6: Fractional function.
Lesson 7: Exponential function.
Lesson 8: Revision and unit test.

Unit 4: Limits.

Lesson 1: Introduction to Limits
Lesson 2: Finding the limit of a function at a point.
Lesson 3: Finding the limit of a function at a point (cont.).
Lesson 4: Finding the limit of a function at infinity.
Lesson 5: Existence of limit of a function at a point.
Lesson 6: Continuity
Lesson 7: Continuity (cont.)
Lesson 8: Revision and unit test.

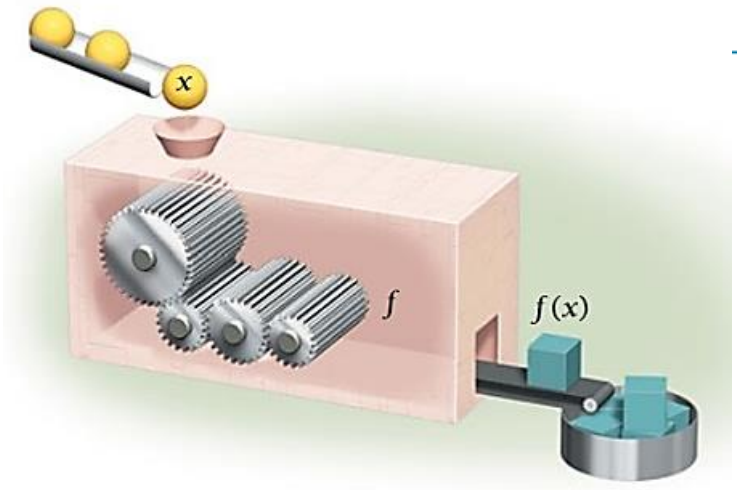
Functions

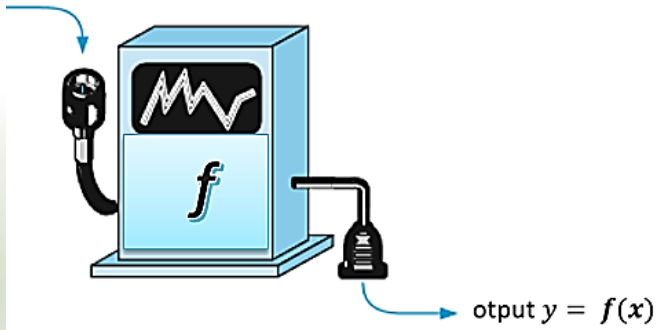
A **relation** from set X to set Y is said to be **function** if **each** element of X has **only** one projection in Y in one of the ordered pairs of the relation.

, it is written as $f: X \rightarrow Y$ and it is read as f is a function from X to Y

Notes:

- Set X is called the **domain** of the function f
- Set Y is called the co-domain of the function f
- Set of images of elements of X in Y is called the **range** of the function f
- If f is a function from **X** to itself, we say that f is a function on **X**.





Example (1)

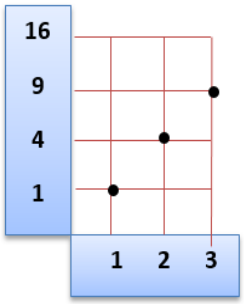
If **X** = {1, 2, 3}, **Y** = {1, 4, 9,16} and R is a relation from **X** to **Y** where a R b means: $a^2 = b$ for each a ∈ **X**, b ∈ **Y**, write R and represent it by Cartesian diagram. Is R a function? Why? Mention the range of f if exists

Solution

R = {(1,1), (2,4),(3,9)}

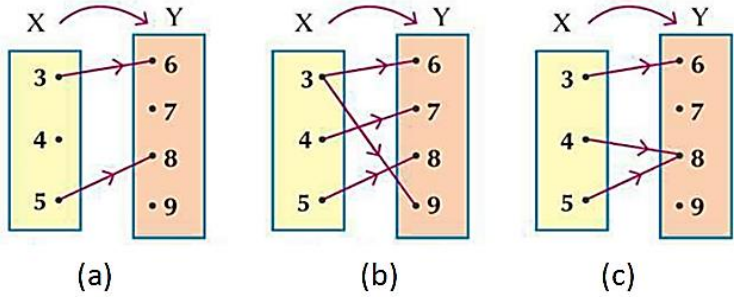
R is a function because each element in X has only one image in Y

The range = {1,4,9}



Example (2)

Which of the shown relations represents a function from X to Y?



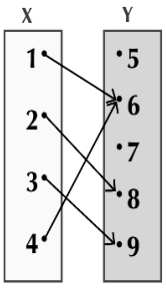
Solution

(a) Not function because the element 4 ∈ X, has no image in Y
(b) Not function because the element 3 ∈ X, has two images in Y
(c) Function because each element ∈ X, has only one image in Y

Example (3)

The opposite figure represents the arrow diagram of a function ***f*, complete:**

1) ***f*** **=** .....
2) The domain of the function ***f*** is .....
3) The co-domain of the function is .....
4) The range of the function is .....



Solution

1) ***f*** **=** {(1,6), (2,8),(3,9),(4,6)}
2) The domain of the function ***f*** is X = {1,2,3,4}
3) The co-domain of the function is Y = {5,6,7,8,9}
4) The range of the function is {6,8,9}

The real functions

A function ***f*** is called a real function if each of its domain and its co-domain is the set of real numbers **R** or a subset of it.

If $f : X \rightarrow Y$ is a function from a non-empty set X to another non-empty set Y, where X ⊂ R , Y ⊂ R (R: set of all real numbers), then we say that *f* is a real function

Polynomial functions

The function f : $f(x) = a_0 + a_1x + a_2x^2 + \ldots\ldots a_nx^n$

Where: a_0 , a_1 , a_2 , ……$a_n \in N$ is called a **polynomial** function of degree n

For example

- ***f(x) = 2x – 3*** is a polynomial function of degree 1 or 1st degree
- ***f(x) = 4x² + 5x – 3*** is a polynomial function of degree 2 or 2nd degree
- ***f(x) = x³ + 2x – 3*** is a polynomial function of degree 3 or 3rd degree
- ***f(x) = 3*** is a polynomial function of degree 0 or zero degree

Example (4)

If $f(x) = 2x - 3$ and $g(x) = x^2 - 1$, then find each of the following:

(1) $f(2)$ (2) $g(2)$

(3) $f(\frac{1}{2})$ (4) $g(1)$

(5) $f(0)$ (6) $g(\sqrt{3})$

(7) If $f(m) = 7$, then find m (8) If $g(m) = 8$, then find m

Solution

(1) $f(2) = 2(2) - 3 = 1$

(2) $g(2) = (2)^2 - 1 = 3$

(3) $f\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right) - 3 = -2$

(4) $g(1) = (1)^2 - 1 = 0$

(5) $f(0) = 2(0) - 3 = -3$

(6) $g(\sqrt{3}) = (\sqrt{3})^2 - 1 = 2$

(7) If $f(m) = 7$, then find m

$2m - 3 = 7 \therefore 2m = 10 \therefore m = 5$

(8) If $g(m) = 8$, then find m

$m^2 - 1 = 8 \therefore m^2 = 9 \therefore m = \pm 3$

Example (5)

Determine the degree of each of the following functions:

(a) $f(x) = 2x^3 - 5$ (b) $f(x) = x(2x - 3) - 2x^2$

(b) $f(x) = 2 - x^4$ (d) $f(x) = \frac{1}{2}$

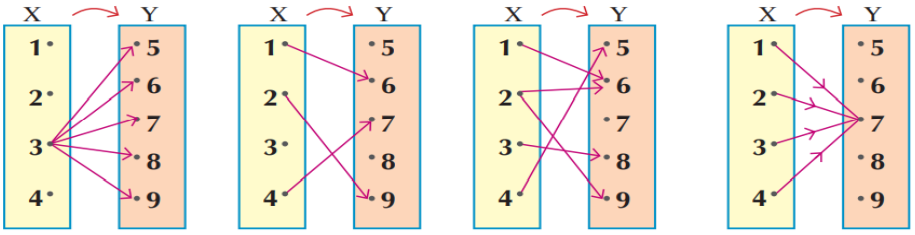
Solution

(a) $f(x) = 2x^3 - 5$ is a polynomial function of 3rd degree
(b) $f(x) = x(2x - 3) - 2x^2 = 2x^2 - 3x - 2x^2 = -3x$ is a function of 1st degree
(c) $f(x) = 2 - x^4$ is a polynomial function of 4th degree
(d) $f(x) = \frac{1}{2}$ is a polynomial function of zero degree

Exercise (3-1)

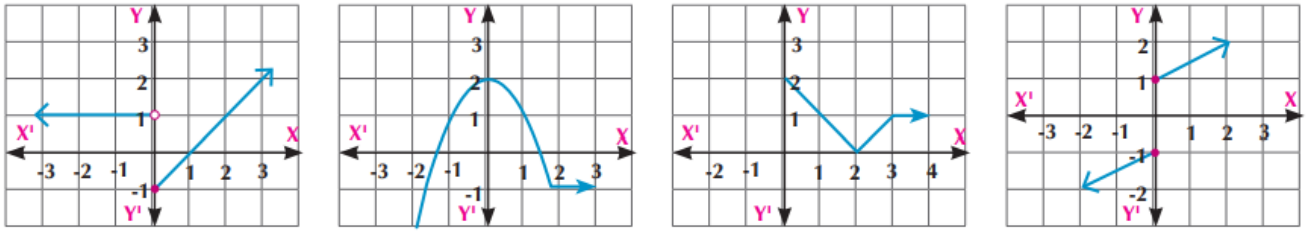
1-Which of the relations shown represents a function from X to Y?

(a) (b) (c) (d)



2-Which of the figures shown does not represent a function? Why?

(a) (b) (c) (d)



3- The relation shown by the set of the ordered pairs and doesn't represent a function is:

(a) R_1 ={(1, 3), (3, 5), (5, 7), (7, 9)}

(b) R_2 = {(2, 3), (3, 4), (2, 1), (3, 5)}

(c) R_3 ={(0, 3), (1, 3), (2, 3), (3, 3)}

(d) R_4 ={(-3, 5), (-1, 5), (0, 5), (2, 5)}

4- Which of the following relations does not represent y as a function of x:

(a) $y = 3x + 1$

(b) $y = x^2 - 1$

(c) $x = y^2 - 4$

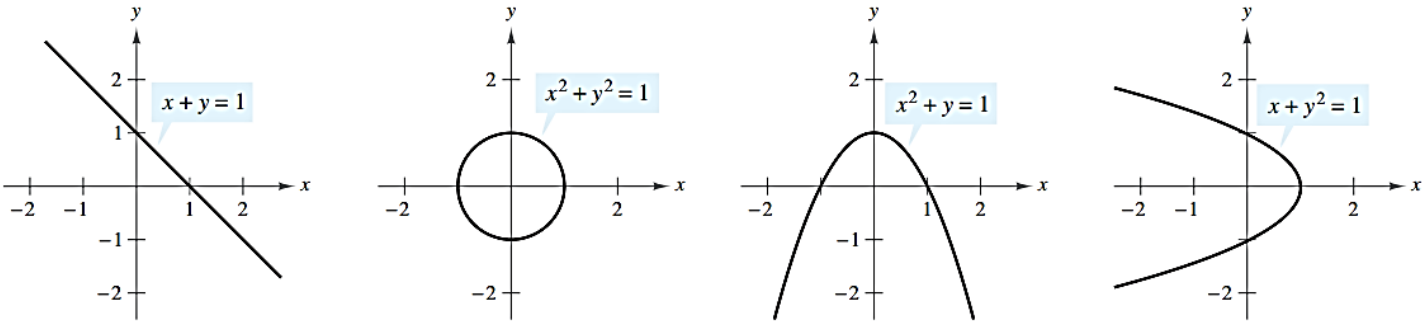
(d) $y = \sin x$

5- if $f(x) = 3x + 1$ and $g(x) = x^2 + 2$, then find each of the following:

(1) $f(1)$ (2) $g(1)$ (3) $f(\frac{1}{3})$ (4) $g(-1)$ (5) $f(0)$

(6) $g(\sqrt{3})$ (7) If $f(m) = 13$, then find m (8) If $g(m) = 27$, then find m

6-Which of the figures shown does not represent a function? Why?



The **Domain** of the function $y = f(x)$ is the set of **all possible values of x**

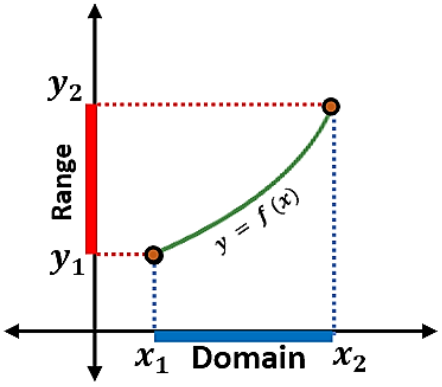
The **Range** of the function $y = f(x)$ is the set of **all possible values of y**

The figure represents the curve of a function $y = f(x)$

From the graph we deduce that:

The Domain = $[x_1, x_2]$

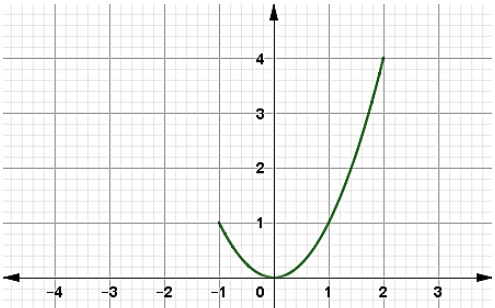
The Range = $[y_1, y_2]$



The **domain** may be mentioned beside the rule of the function for example:

$f(x) = x^2, -1 \leq x \leq 2$

Domain $= [-1,2]$, Range =[0,4]

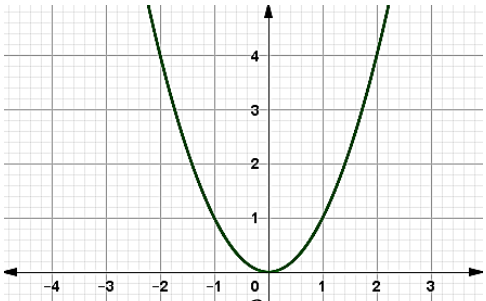


$f(x) = x^2, x \leq 0$

Domain $=]-\infty, 0]$, Range=[0,∞[

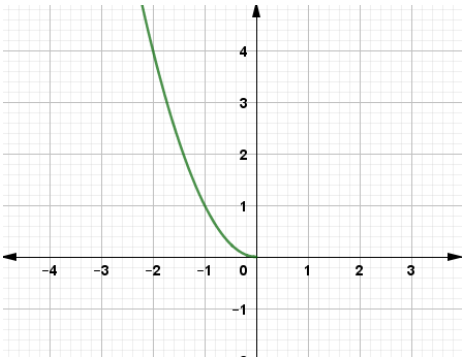
$f(x) = x^2$

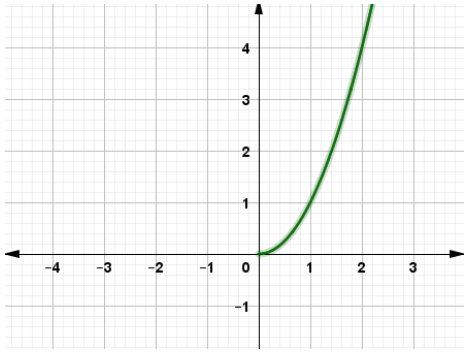
Domain $=$ R , Range=[0,∞[



$f(x) = x^2, x > 0$

Domain $= [0, \infty[$, Range=[0,∞[





If the domain is not mentioned, we can deduce it from the rule of the function as following:

The domain of a **Polynomial functions** is R

- $f(x) = x^2 + 3$
- $f(x) = x^3 - 2x + 5$
- $f(x) = 3x + 1$
- $f(x) = 2$
- $f(x) = 11 - 3x^4$
- $f(x) = 3x^5 - 2x^2 - 1$

In each of the function above we can substitute x by any number, so the **domain** is **R**

The domain of the square root function $\boldsymbol{y} = \sqrt{\boldsymbol{g}(\boldsymbol{x})}$ must satisfy the condition $\boldsymbol{g}(\boldsymbol{x}) \geq 0$

For example, if $f(x) = \sqrt{\boldsymbol{x} - \mathbf{3}}$ can we find $f(3)?, f(4)? f(0)? f(2)? f(-1)?$

$f(3) = \sqrt{3-3} = \sqrt{0} = 0 \in$ R ✓

$f(4) = \sqrt{4-3} = \sqrt{1} = 1 \in$ R ✓

$f(0) = \sqrt{0-3} = \sqrt{-3} \notin$ R ✗

$f(2) = \sqrt{2-3} = \sqrt{-1} \notin$ R ✗

$f(-1) = \sqrt{-1-3} = \sqrt{-4} \notin$ R ✗

Example (1)

Determine the domain of each of the following functions:

(a) $f(\mathrm{x}) = 4\mathrm{x}^3 - 5\mathrm{x} + 2$ (b) $f(\mathrm{x}) = \sqrt{\boldsymbol{x} - \mathbf{1}}$

Solution

(a) $\boldsymbol{f}(\mathbf{x}) = \mathbf{4}\boldsymbol{x}^{\mathbf{3}} - \mathbf{5}\boldsymbol{x} + \mathbf{2}$

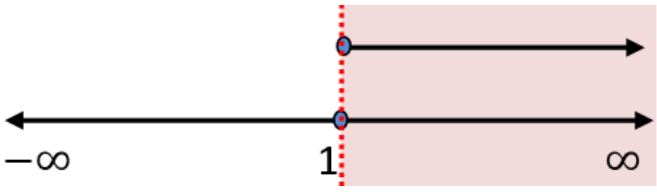
$f(\mathrm{x}) = 4\mathrm{x}^3 - 5\mathrm{x} + 2$ is a polynomial function ,then the domain = R

(b) $\boldsymbol{f}(\mathbf{x}) = \sqrt{\boldsymbol{x} - \mathbf{1}}$

$x - 1 \geq 0$

$x \geq 1$

$\therefore x \in [1, \infty[$



$\therefore$ The domain $= [1, \infty[$

Example (2)

Determine the domain of each of the following functions:

(b) $f(\mathrm{x}) = \sqrt{\mathbf{3} - \boldsymbol{x}}$ (b) $f(\mathrm{x}) = \sqrt{\boldsymbol{x} + \mathbf{2}}$ (c) $f(\mathrm{x}) = \sqrt{\mathbf{2}\boldsymbol{x} - \mathbf{3}}$

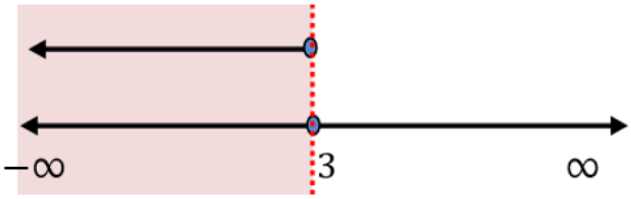
Solution

(a) $\boldsymbol{f}(\mathbf{x}) = \sqrt{\mathbf{3} - \boldsymbol{x}}$

$3 - x \geq 0$

$-x \geq -3\,,\ x \leq 3$

$\therefore x \in]-\infty, 3]$

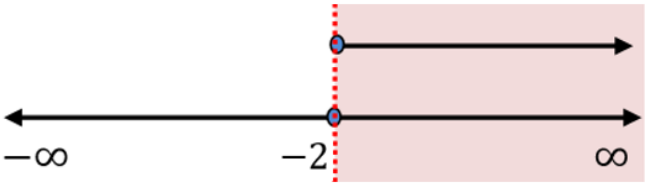


$\therefore$ The domain $=]-\infty, 3]$

(b) $\boldsymbol{f}(\mathbf{x}) = \sqrt{\boldsymbol{x} + \mathbf{2}}$

$x + 2 \geq 0$

$x \geq -2 \quad \therefore x \in [-2, \infty[$

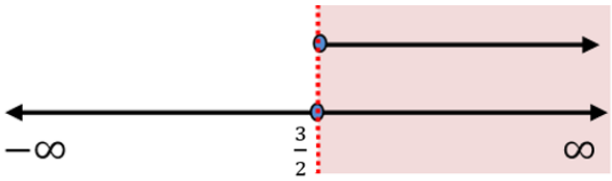


$\therefore$ The domain $= [-2, \infty[$

(c) $\boldsymbol{f}(\mathbf{x}) = \sqrt{\mathbf{2}\boldsymbol{x} - \mathbf{3}}$

$2x - 3 \geq 0\,,\ \ 2x \geq 3\ \ ,\ x \geq \frac{3}{2}$

$\therefore x \in [\frac{3}{2}, \infty[$



$\therefore$ The domain $= [-2, \infty[$

The domain of the fractional function $\boldsymbol{y} = \frac{1}{\boldsymbol{g}(\boldsymbol{x})}$ is R− *the values that makes* $\boldsymbol{g}(\boldsymbol{x}) = 0$

Example (3)

Determine the domain of each of the following functions:

(c) $f(\mathrm{x}) = \frac{1}{x-3}$ (b) $f(\mathrm{x}) = \frac{1}{x^2-4}$ (c) $f(\mathrm{x}) = \frac{1}{x^2+1}$

Solution

(a) $f(\mathrm{x}) = \frac{1}{x-3}$

The denominator $x - 3 = 0$ at $x = 3$,then the domain $= R - \{3\}$

(b) $f(\mathrm{x}) = \frac{1}{x^2-4}$

The denominator $x^2 - 4 = 0 \therefore (x-2)(x+2) = 0 \therefore x = 2 \; or \; x = -2$, then the domain $= R - \{2, -2\}$

(c) $f(\mathrm{x}) = \frac{1}{x^2+1}$

The denominator: $x^2 + 1 = 0$, there is no real value of x makes $x^2 + 1 = 0$, then the domain $= R$

Exercise (3-2)

Determine the domain of each of the following functions:

(a) $f(\mathrm{x}) = \mathrm{x}^4 - 5\mathrm{x} + 2$

(b) $f(\mathrm{x}) = \sqrt{x - 3}$

(c) $f(\mathrm{x}) = \sqrt{x + 2}$

(d) $f(\mathrm{x}) = \sqrt{5 - x}$

(e) $f(\mathrm{x}) = \sqrt{2x - 1}$

(f) $f(\mathrm{x}) = \sqrt{6 - 3x}$

(g) $f(\mathrm{x}) = \sqrt{\mathrm{x}^2 - 4}$

(h) $f(\mathrm{x}) = \sqrt{9 - \mathrm{x}^2}$

(i) $f(\mathrm{x}) = \sqrt{\mathrm{x}^2 + 1}$

(j) $f(\mathrm{x}) = 2x$

(k) $f(\mathrm{x}) = \frac{x}{2}$

(l) $f(\mathrm{x}) = \frac{2}{x}$

(m) $f(\mathrm{x}) = \frac{1}{x-5}$

(n) $f(\mathrm{x}) = \frac{1}{x^2-9}$

(o) $f(\mathrm{x}) = \frac{1}{x^2+4}$

Unit 3	Lesson 3	Constant function and linear function

The constant function

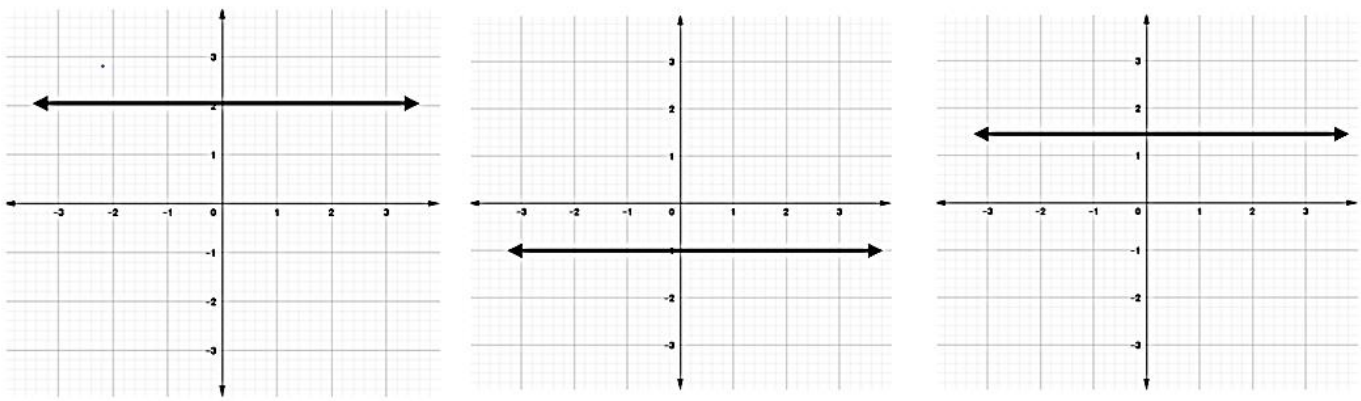
The function $\boldsymbol{f}: f(x) = c$, $c \in R$ is called a constant function , and is represented graphically by a straight line parallel to x-axis and cuts y axis at point (o, c)

Example (1)

Represent each of the following functions:

(a) $f(x) = 2$ (b) $f(x) = -1$ (c) $f(x) = 1.5$

Solution (a) (b) (c)

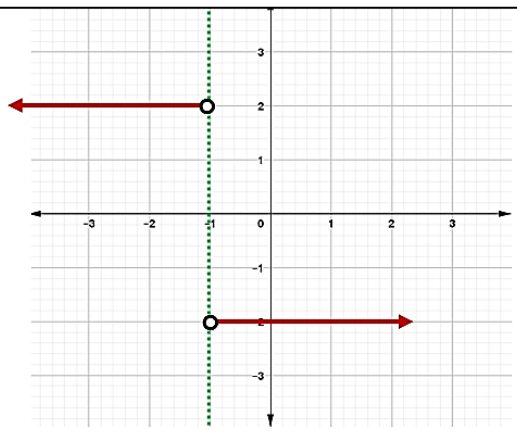


Example (2)

Represent the function $f: f(x) = \begin{cases} -2 & , x > -1 \\ 2 & , x < -1 \end{cases}$

$f(x) = -2$ is a constant function , so we represent it by a ray parallel to the x-axis and cuts y axis at point $(0, -2)$, but here we have a condition $x > -1$, that's why we draw only the part on $]-1, \infty[$

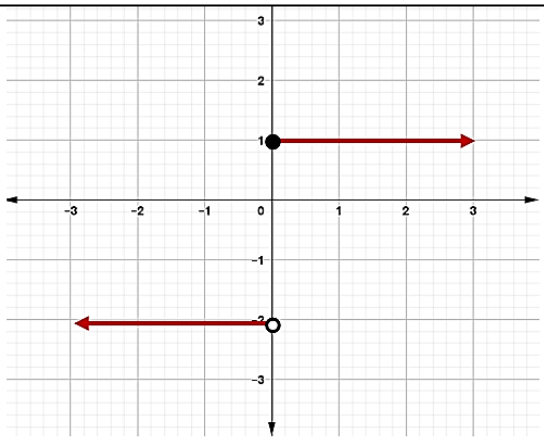
Also $f(x) = 2$ is a constant function , so we represent it by a ray parallel to the x-axis and here we have a condition $x < -1$, that's why we draw only the part on $]-\infty, -1[$



Notes : Functios having more than one base are called piecwise functions.

Example (3)
1

Represent the function $f: f(x) = \begin{cases} 1 & , x \geq 0 \\ -2 & , x < 0 \end{cases}$



$f(x) = 1$ is a constant function , so we represent it by a ray parallel to the x-axis and cuts y axis at pont $(0,1)$, but here we have a condition $x \geq 0$,that's why we draw only the part on $[0, \infty[$
Also $f(x) = -2$ is a constant function , so we represent it by a ray parallel to x-axis and here we have a condition $x < 0$
,that's why we draw only the part on $]-\infty, 0[$

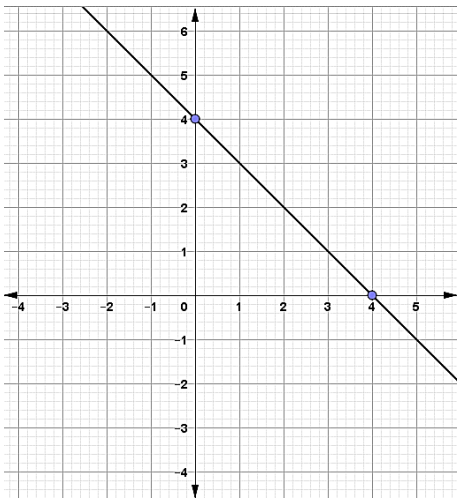
The linear function

The function $\boldsymbol{f}$: $f(x) = bx + c$ where $b \neq 0$ is called a linear function (1st degree function) and it is represented graphically by a straight line cuts the x-axis at point $(\frac{-c}{b}, 0)$, and cuts the y-axis at $(0, c)$

Example (4)

Represent the function $f: f(x) = 4 - x$

Solution



1st method

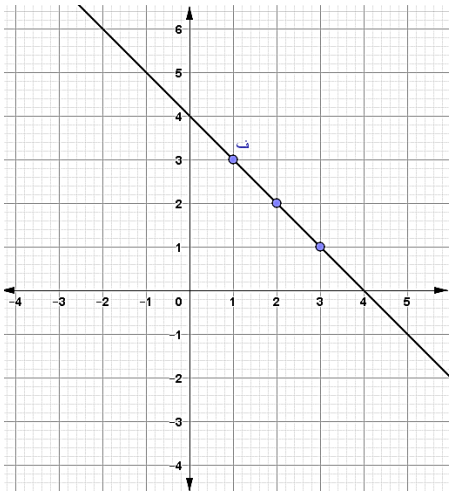
Using the intercepted parts from the x-axis and the y-axis So, we look for the two points: (0 , ...) and (....., 0)
Put x = 0 $\therefore y = f(0) = 4 - 0 \therefore y = 4$
Put $f(x) = 0 \therefore 4 - x = 0 \therefore x = 4$
We have the two points $(0, 4)$ and $(4,0)$

2nd method

Using any three points, choose any three values of x , then find the corresponding values of $y = f(x) = 4 - x$

x	1	2	3
$y = f(x)$	3	2	1

Two points only are sufficient, but we take the third point to make sure that we get the right line



Example (5)

Represent the function $f: f(x) = \frac{2}{3}x - 2$

Solution

1st method

Put x = 0$\therefore y = f(0) = \frac{2}{3}(0) - 2 \therefore y = -2$

Put $f(\mathrm{x}) = 0 \ \therefore \frac{2}{3}x - 2 = 0$

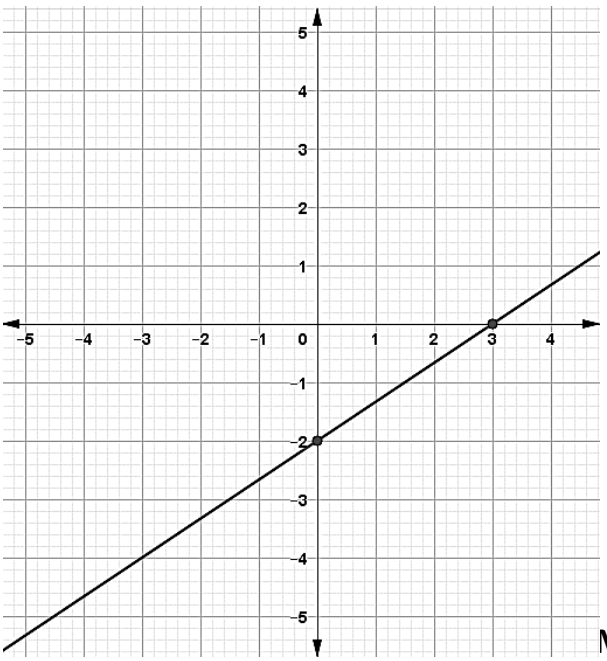
$\therefore x = 3 \therefore \frac{2}{3}x = 2$

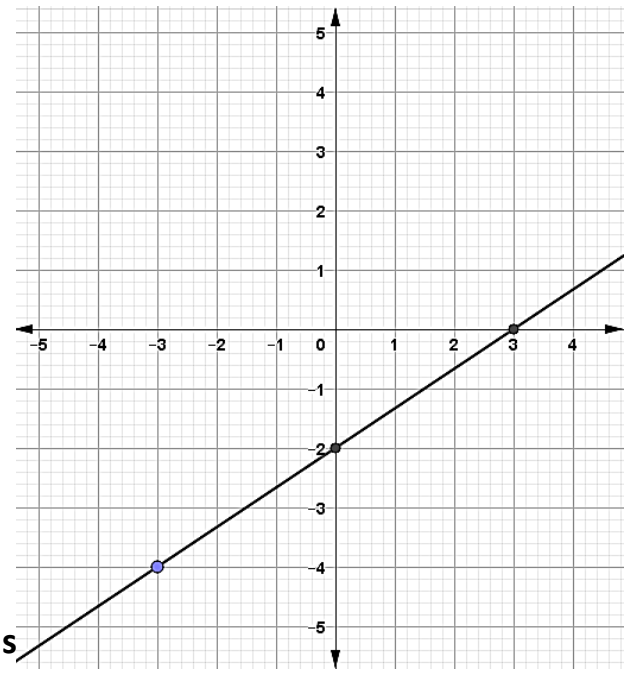
Now we have the two points (0 , -2), (3, 0)

2nd method

$f(x) = \frac{2}{3}x - 2$, Now we choose any three values of x as – 3 , 0 ,3

x	– 3	0	3
$y = f(x)$	– 4	– 2	0



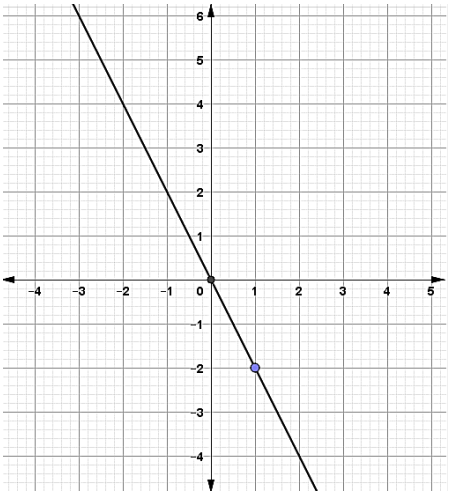


Example (6)

Represent the function $f: f(x) = -2x$

Solution

Put x = 0 $\therefore\ y = f(0) = -2(0) = 0 \therefore\ y = 0$
point $(0,0)$
The straight line passes through the origin
We need another point
Put x = 1 $\therefore\ y = f(1) = -2(1) = -2$
, point $(1, -2)$
or you may use the other method



Exercise (3-3)

1- Represent graphically each of the following linear functions:

a) $f(x) = 3x - 2$

b) $f(x) = 5 - x$

c) $f(x) = 2x$

d) $f(x) = \frac{1}{2}x$

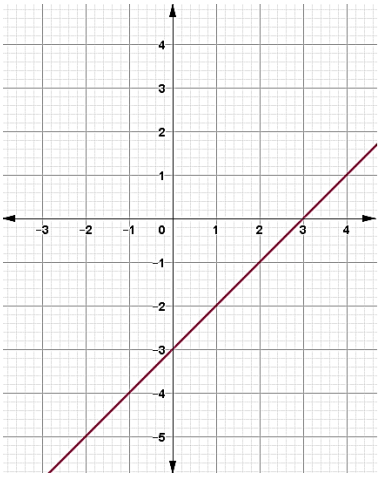
e) $f(x) = x + 2$

f) $f(x) = 2x + 3$

g) $f(x) = 6 - 3x$

h) $f(x) = -2x$

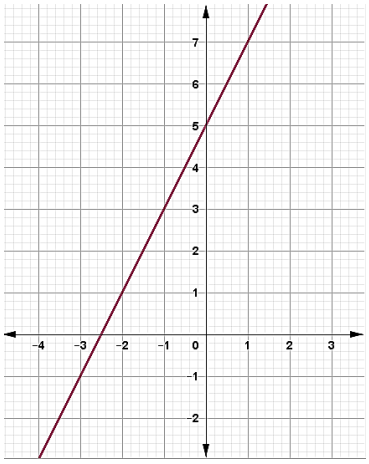
2- Choose the correct answer:



i) The opposite figure represents the graph of

the function $f(x) =$ …….

(a) $x - 3$
(b) $x + 3$
(c) $3 - x$
(d) $3x$



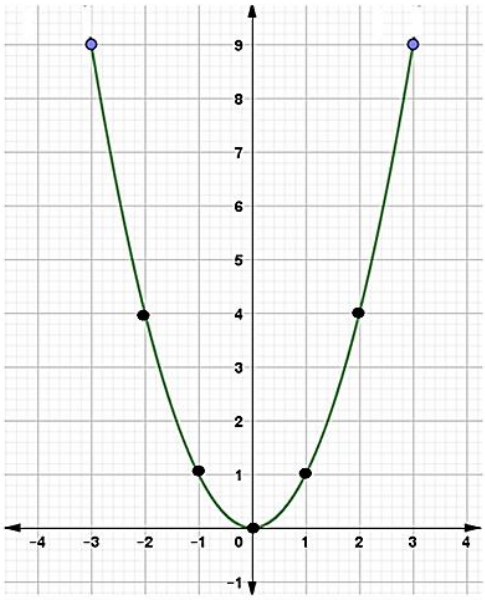
ii)The opposite figure represents the graph of

the function $f(x) =$ …….

(e) $2x - 5$
(f) $2x + 5$
(g) $5x - 3$
(h) $5x + 2$

The quadratic function

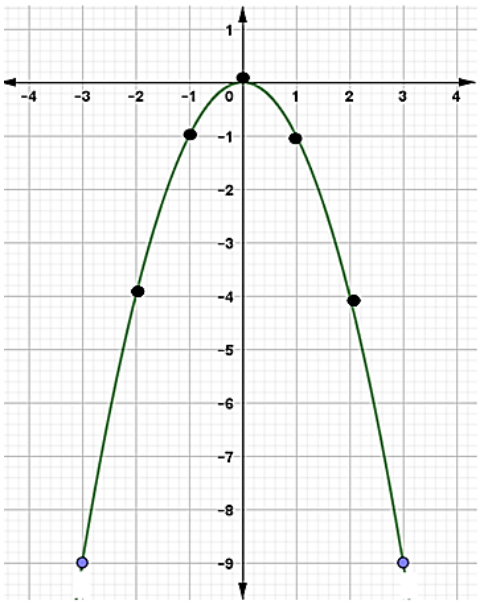
The function $\boldsymbol{f}$: $f(x) = ax^2 + bx + c$
where a , b and c are real numbers , $a \neq 0$
is called a quadratic function and it is a polunomial function of the second degree.
The graph of the quadratic function:
$f : f(x) = x^2$. , $x \in [-3 , 3]$



x	-3	-2	-1	0	1	2	3
(x)	9	4	1	0	1	4	9

- The curve is symmetric about the y-axis whose equation is X = 0
- The point (0 ,0) is the point of the vertex of the curve.
- The minimum value of the function is zero

The graph of the quadratic function:
f : $f(x) = x^2, x \in [-3 , 3]$



x	-3	-2	-1	0	1	2	3
f(x)	-9	-4	-1	0	-1	-4	-9

- The curve is symmetric about the y-axis whose equation is X = 0
- The point (0 ,0) is the point of the vertex of the curve.
- The maximum value of the function is zero

The vertex of the quadratic function f : $f(x)ax^2 + bx + c$ can be obtained from the relation :

- The vertex $V(\frac{-b}{a}, f(\frac{-b}{a}))$
- Equation of line of symmetry: $x = \frac{-b}{a}$
- Minmum or Maximum value : $y = f(\frac{-b}{a})$

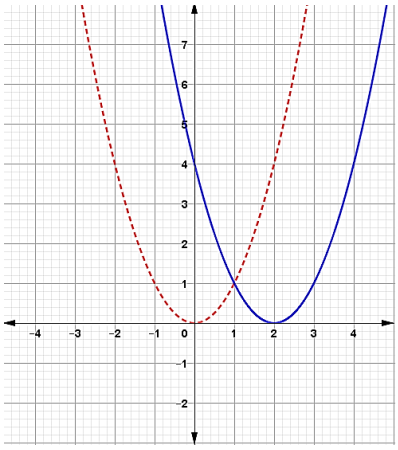
Example (1)

Use the curve of the function f where $f(x) = x^2$ to represent the functions g where :

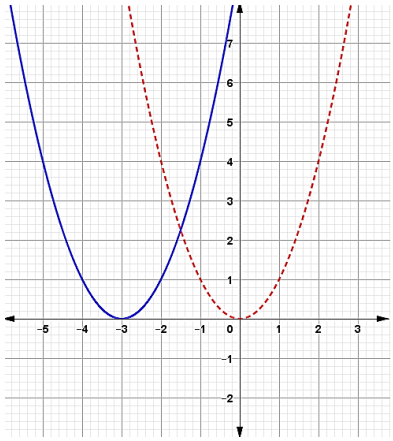
a) $g(x) = (x-2)^2$ b) $g(x) = (x+3)^2$

c) $g(x) = x^2 - 2$ d) $g(x) = x^2$+3

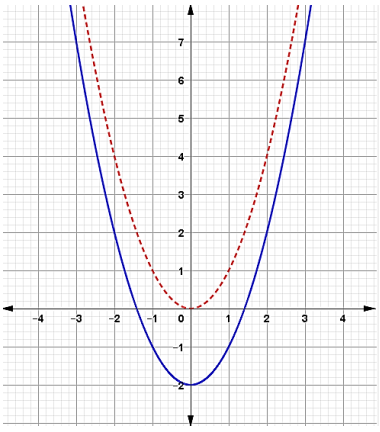
Solution



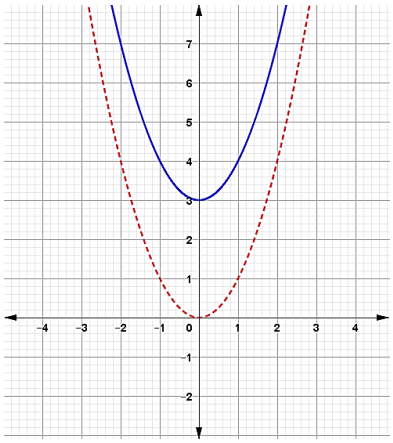
The curve of $g(x) = (x-2)^2$is the same curve of $f(x) = x^2$ by translation 2 units in the positive direction of x –axis and the curve vertex point is (2, 0).



The curve of $g(x) = (x+3)^2$is the same curve of $f(x) = x^2$ by translation 3 units in the negative direction of x –axis and the curve vertex point is $(-3, 0)$.



The curve of $g(x) = \mathrm{x}^2 - 2$ is the same curve of $f(x) = x^2$ by translation 2 units in the negative direction of y –axis and the curve vertex point is $(0, -2)$.



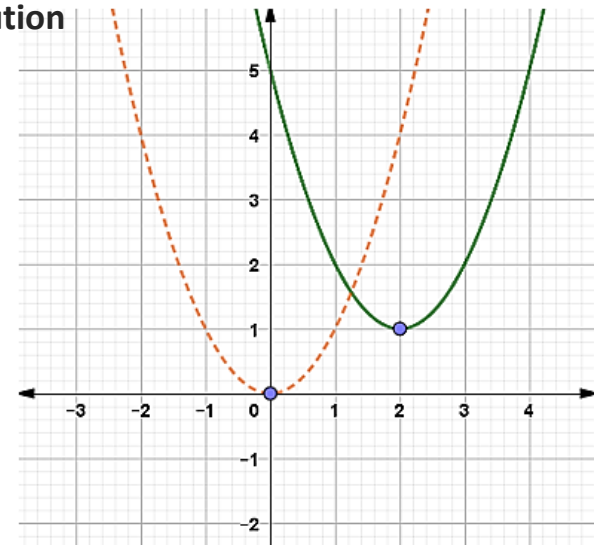
The curve of $g(x) = \mathrm{x}^2 + 3$ is the same curve of $f(x) = x^2$ by translation 3 units in the positive direction of y –axis and the curve vertex point is $(0, 3)$.

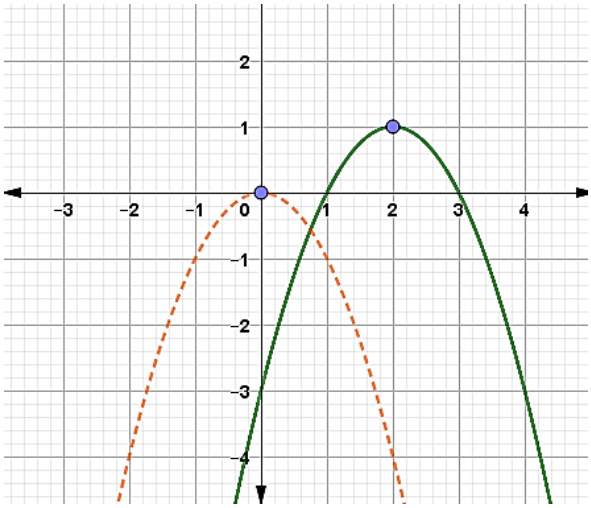
Example (2)

Use the curve of the function f where $f(x) = x^2$ to represent the functions g where :

a) $g(x) = (x-2)^2 + 1$ b) $g(x) = -(x-2)^2+1$

Solution





The curve of $g(x) = (x-2)^2 + 1$ is the same curve of $f(x) = x^2$ by translation 2 units in the positive direction of the x –axis and 1 units in the positive direction of the y –axis and the curve vertex point is (2, 1).

The curve of $g(x) = -(x-2)^2 + 1$is the same curve of $f(x) = -x^2$ by translation 2 units in the positive direction of the x –axis and 1 units in the positive direction of The y –axis and the curve vertex point is (2, 1).

Notice that the curve of the quadratic function opens up when the qcoifficent of x^2 is positive and opens down when the qcoifficent of x^2 is negative

Example (3)

Choose the correct answer:

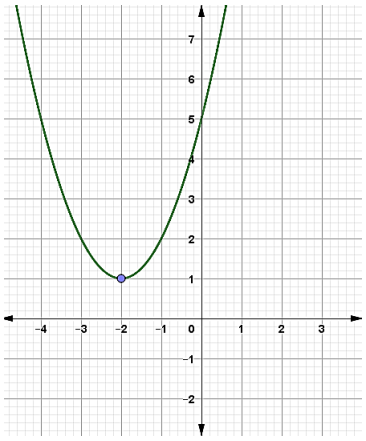
The opposite figure represents the curve of the function f: $f(x) = \ldots\ldots\ldots$

(a) $f(x) = (x-2)^2 + 1$

(b) $f(x) = (x-2)^2 - 1$

(c) $f(x) = (x+2)^2 + 1$

(d) $f(x) = (x+2)^2 - 1$



Solution

(c) $f(x) = (x+2)^2 + 1$

Example (4)

Choose the correct answer:

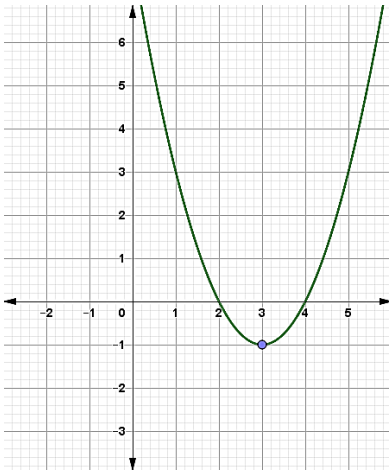
The opposite figure represents the curve of the function f: $f(x) = \ldots\ldots\ldots$

(a) $f(x) = (x-3)^2 + 1$

(b) $f(x) = (x-3)^2 - 1$

(c) $f(x) = (x+3)^2 + 1$

(d) $f(x) = (x+3)^2 - 1$



Solution

(b) $f(x) = (x-3)^2 - 1$

Example (5)

Choose the correct answer:

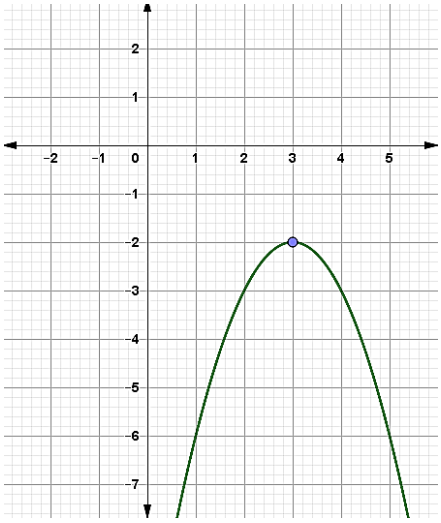
The opposite figure represents the curve of the function f: $f(x) = \ldots\ldots\ldots$

(a) $f(x) = (x-3)^2 + 2$

(b) $f(x) = (x-3)^2 - 2$

(c) $f(x) = -(x-3)^2 + 2$

(d) $f(x) = -(x-3)^2 - 2$



Solution

(d) $f(x) = -(x-3)^2 - 2$

Exercise (3-4)

1- Write the symmetric point of each of the following quadratic functions:

a) $f(x) = (x-1)^2$

b) $f(x) = (x+4)^2$

c) $f(x) = x^2 - 3$

d) $f(x) = x^2 + 1$

e) $f(x) = (x+1)^2 - 2$

e) $f(x) = (x-3)^2 + 1$

g) $f(x) = (x-2)^2 - 2$

h) $f(x) = 3 - x^2$

i) $f(x) = 1 - (x-1)^2$

j) $f(x) = x^2 - 4x + 2$

k) $f(x) = x^2 - 2x + 3$

l) $f(x) = x^2 + 6 + 5$

2- Graph each of the following quadratic functions:-

a) $f(x) = x^2 + 2x + 1, x \in [-4,2]$

b) $f(x) = x^2 - 2,\ x \in [-3,3]$

c) $f(x) = (x - 2)^2 x \in [-1,5]$

d) $f(x) = 3 - 2x - x^2,\ x \in [-4,2]$

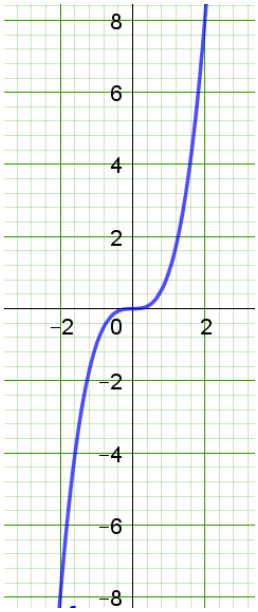
e) $f(x) = 3 - x^2$

f) $(x) = 1 - (x-2)^2$

The simplest form of the **cubic function** is $f : f(x) = x^3$

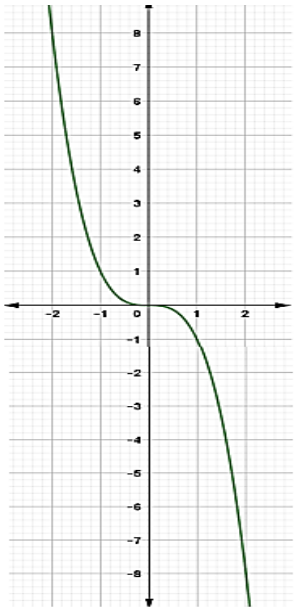
$f : f(x) = x^3$, x ∈ [–2 , 2]

x	-2	-1	0	1	2
(x)	-8	-1	0	1	8



$f : f(x) = -x^3$, x ∈ [–2 , 2]

x	-2	-1	0	1	2
(x)	8	1	0	-1	-8



- The curve is symmetric about the orgion point.

The symmetric point of the graph of the function $f : f(x) = x^3$ is the origion point (0, 0) and the curve lies on the 1st and the 3rd quadrants.

The symmetric point of the graph of the function $f : f(x) = -x^3$ is the origion point (0, 0)) and the curve lies on the 2nd and the 4th quadrants.

The symmetric point of the graph of the function $f : f(x) = x^3 + 2$ is the point (0, 2)

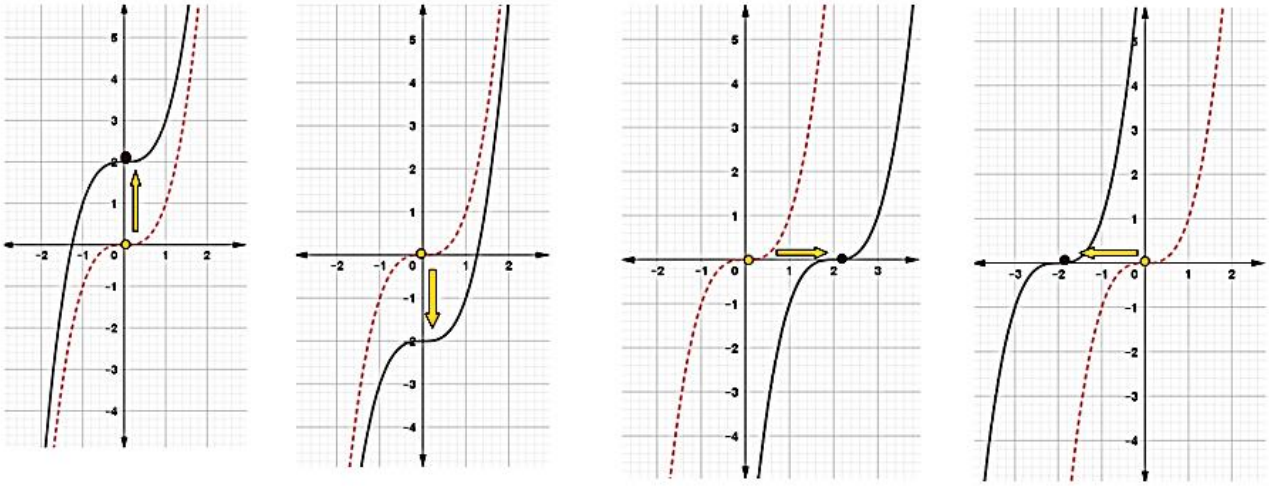
The symmetric point of the graph of the function $f : f(x) = x^3 - 2$ is the point (0, -2)

The symmetric point of the graph of the function $f : f(x) = (x - 2)^3$ is the point (2, 0)

The symmetric point of the graph of the function $f : f(x) = (x + 2)^3$ is the point (-2, 0)

Notict the graph of the following cubic functions

$f(x) = x^3 + 2$ $\quad$ $f(x) = x^3 - 2$ $\quad$ $f(x) = (x-2)^3$ $\quad$ $f(x) = (x+2)^3$



The symmetric point of the graph of the function $f : f(x) = (x-a)^3 + b$ is the point (a, b) and the curve lies on the 1st and the 3rd quadrant.

The symmetric point of the graph of the function $f : f(x) = -(x-a)^3 + b$ is the point (a, b) and the curve lies on the 2nd and the 4th quadrant.

Example (1)

Write the summetric point of the graph of each of the the following functions:

a) $f(x) = (x-1)^3$ $\quad$ b) $f(x) = (x+3)^3$

c) $f(x) = x^3 - 4$ $\quad$ d) $f(x) = x^3+3$

e) $f(x) = (x+1)^3 - 2$ $\quad$ f) $f(x) = 1 - (x+3)^3$

g) $f(x) = -x^3 + 2$ $\quad$ h) $f(x) = -(x-1)^3 - 1$

Solution

a) $f(x) = (x-1)^3$, the symmetric point is $(1, 0)$

b) $f(x) = (x+3)^3$, the symmetric point is $(-3, 0)$

c) $f(x) = x^3 - 4$, the symmetric point is $(0, -4)$

d) $f(x) = x^3+3$, the symmetric point is $(0, 3)$

e) $f(x) = (x+1)^3 - 2$, the symmetric point is $(-1, -2)$

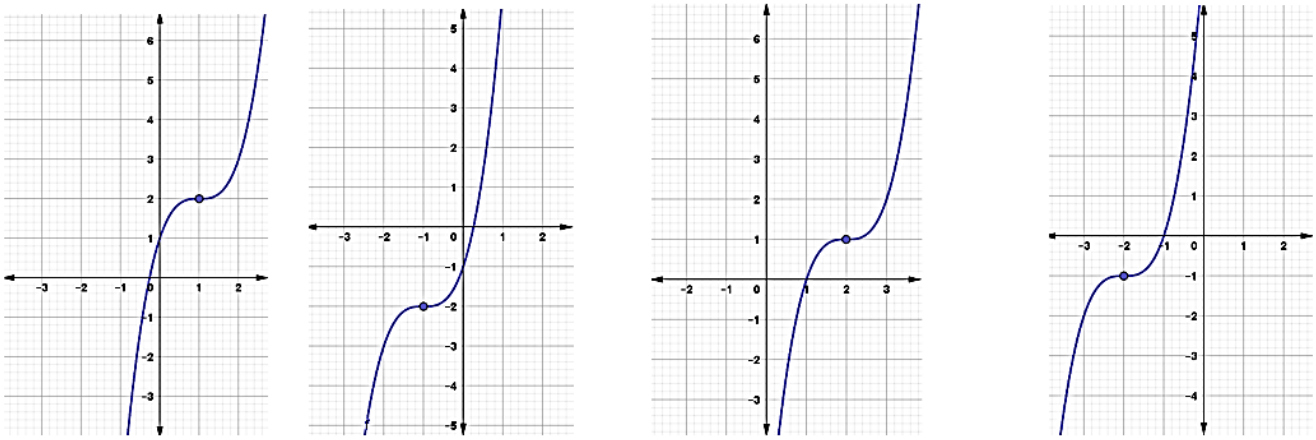
f) $f(x) = 1 - (x+3)^3$, the symmetric point is $(-3, 1)$

g) $f(x) = -x^3 + 2$, the symmetric point is $(0, 2)$

h) $f(x) = -(x-1)^3 - 1$, the symmetric point is $(1, -1)$

Example (2)

Which of the following reprsents the graph of the function $f: f(x) = (x+1)^3 - 2$?



(a) (b) (c) (d)

Solution

the figure that reprsents the graph of the function $f: f(x) = (x+1)^3 - 2$ is figure (b)

Example (3)

Choose the correct answer:

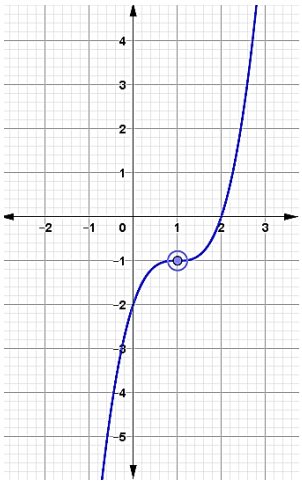
The opposite figure represents the curve of the function $f: f(x) = \ldots\ldots\ldots$

(a) $(x+1)^3 + 1$

(b) $(x+1)^3 - 1$

(c) $(x-1)^3 + 1$

(d) $(x-1)^3 - 1$

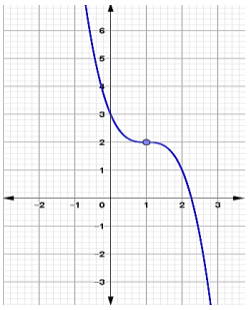


Solution

(d) The opposite figure represents the curve of the function $f: f(x) = (x-1)^3 - 1$

Exercise (3-5)

1- Choose the correct answer:



The opposite figure represents the curve of the function f: $f(x) = \ldots\ldots\ldots$

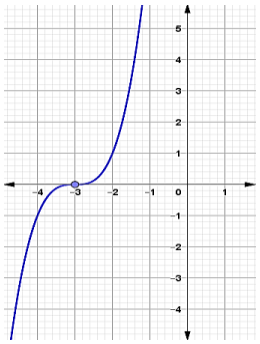
(a) $(x-2)^3+1$

(b) $-(x-2)^3+1$

(c) $(x-1)^3+2$

(d) $-(x-1)^3+2$

2- Choose the correct answer:



The opposite figure represents the curve of the function f: $f(x) = \ldots\ldots\ldots$

(a) $(x-3)^3$

(b) $(x+3)^3$

(c) x^3+3

(d) x^3-3

3- Write the symmetric point of each of the following functions:

a) $f(x) = (x+1)^3$

b) $f(x) = (x+2)^3$

c) $f(x) = x^3-1$

d) $f(x) = x^3+1$

e) $f(x) = (x+7)^3-1$

f) $f(x) = (x-3)^3+2$

g) $f(x) = (x-2)^3-4$

h) $f(x) = 2-x^3$

i) $f(x) = 1-(x+1)^3$

4- Graph each of the following functions:

a) $f(x) = (x-1)^3$

b) $f(x) = x^3-2$

c) $f(x) = -x^3+3$

d) $f(x) = x^3+3$

e) $f(x) = (x+2)^3+1$

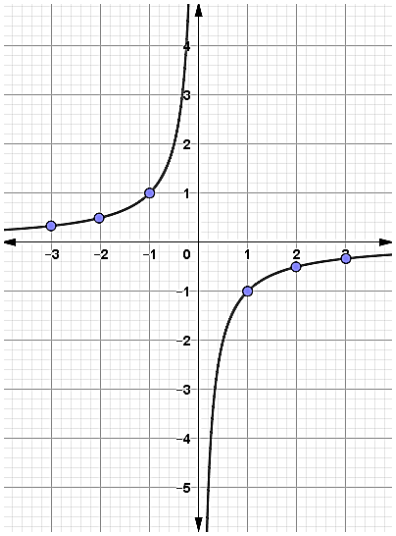
f) $f(x) = (x-1)^3+3$

The simplest form of the **fractional function is**

The function $\boldsymbol{f}$: $f(x) = \frac{1}{x}, x \neq 0$

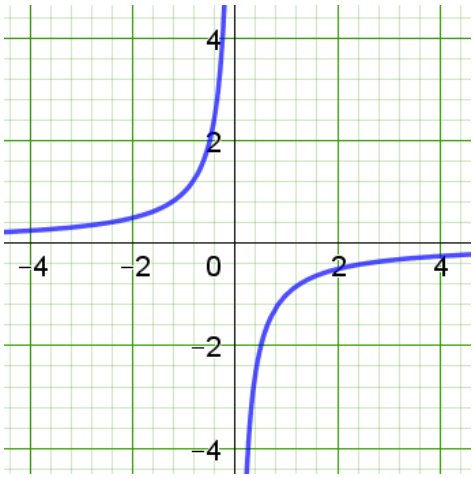
$\boldsymbol{f}$: $f(x) = \frac{1}{x}, x \neq 0$

X	-3	-2	-1	1	2	3
(x)	$\frac{-1}{3}$	$\frac{-1}{2}$	-1	1	$\frac{1}{2}$	$\frac{1}{3}$



$\boldsymbol{f}$: $f(x) = -\frac{1}{x}, x \neq 0$

x	-3	-2	-1	1	2	3
(x)	$\frac{1}{3}$	$\frac{1}{2}$	1	-1	$\frac{-1}{2}$	$\frac{-1}{3}$



The symmetric point of the graph of the function f : f(x) = $\frac{1}{x}$ is the origion point (0, 0) and the curve lies on the 1st and the 3rd quadrants.

The symmetric point of the graph of the function f : f(x) = $-\frac{1}{x}$ is the origion point (0, 0)) and the curve lies on the 2nd and the 4th quadrants.

The symmetric point of the graph of the function f : f(x) = $\frac{1}{x} + 2$ is the point (0, 2)

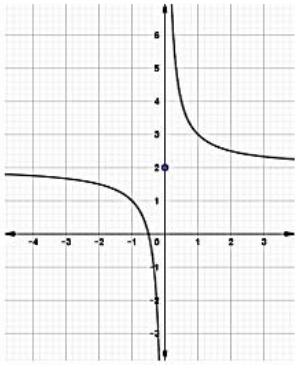
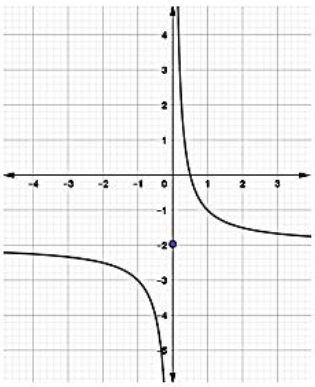
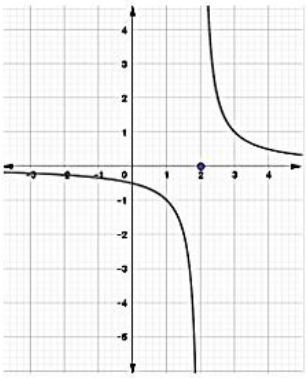
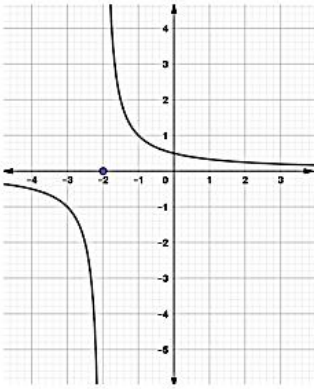
The symmetric point of the graph of the function f : f(x) = $\frac{1}{x} - 2$ is the point (0, -2)

The symmetric point of the graph of the function f : f(x) = $\frac{1}{x-2}$ is the point (2, 0)

The symmetric point of the graph of the function f : f(x) = $\frac{1}{x+2}$ is the point (-2, 0)

Notict the graph of the following cubic functions

$f(x) = \frac{1}{x} + 2$ $\quad$ $f(x) = \frac{1}{x} - 2$ $\quad$ $f(x) = \frac{1}{x-2}$ $\quad$ $f(x) = \frac{1}{x+2}$

Example (1)

Write the symmetric point of the graph of each of the the following functions:

a) $f(x) = \frac{1}{x-1} + 2$ $\quad$ b) $f(x) = \frac{1}{x-2} + 1$

c) $f(x) = \frac{1}{x} + 2$ $\quad$ d) $f(x) = \frac{1}{x} - 2$

e) $f(x) = \frac{1}{x-3}$ $\quad$ f) $f(x) = -\frac{1}{x-1} + 1$

g) $f(x) = \frac{1}{x+1} - 2$ $\quad$ h) $f(x) = 3 - \frac{1}{x-3}$

Solution

a) $f(x) = \frac{1}{x-1} + 2$, the symmetric point is $(1, 2)$

b) $f(x) = \frac{1}{x-2} + 1$, the symmetric point is $(2, 1)$

c) $f(x) = \frac{1}{x} + 2$, the symmetric point is $(0, 2)$

d) $f(x) = \frac{1}{x} - 2$, the symmetric point is $(0, -2)$

e) $f(x) = \frac{1}{x-3}$, the symmetric point is $(3, 0)$

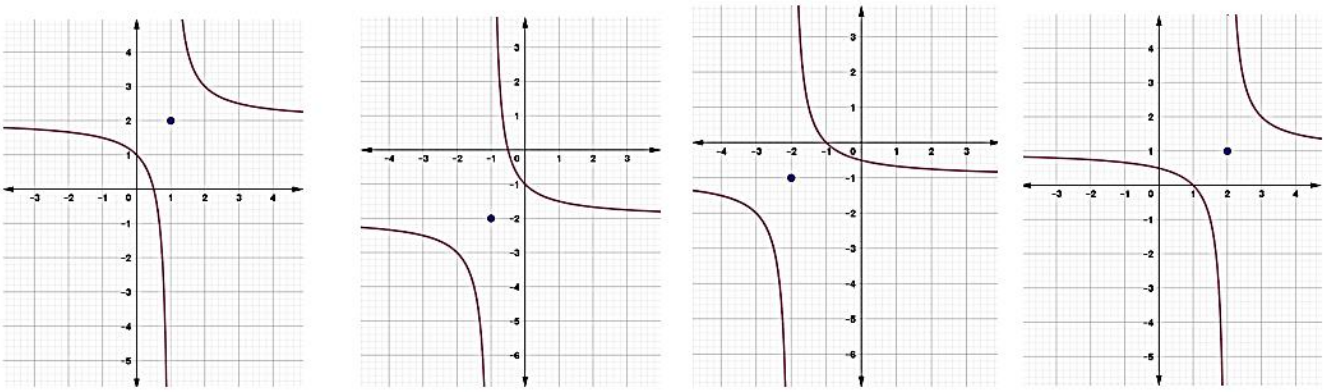
f) $f(x) = -\frac{1}{x-1} + 1$, the symmetric point is $(1, 1)$

g) $f(x) = \frac{1}{x+1} - 2$, the symmetric point is $(-1, -2)$

h) $f(x) = 3 - \frac{1}{x-3}$,the symmetric point is $(3, 3)$

Example (2)

Which of the following reprsents the graph of the function $f: f(x) = \frac{1}{x-1} + 2$?



(a) (b) (c) (d)

Solution

the figure that reprsents the graph of the function $f: f(x) = \frac{1}{x-1} + 2$ is figure (a)

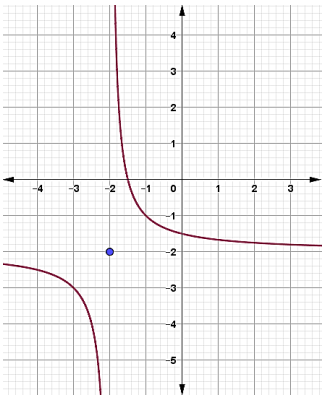
Example (3)

Choose the correct answer:

The opposite figure represents the curve of the function $f: f(x) = \ldots\ldots\ldots$

(a) $\frac{1}{x-2} - 2$ (b) $\frac{1}{x-2} + 2$

(c) $\frac{1}{x+2} + 2$ (d) $\frac{1}{x+2} - 2$



Solution

(d) The opposite figure represents the curve of the function $f: f(x) = \frac{1}{x+2} - 2$

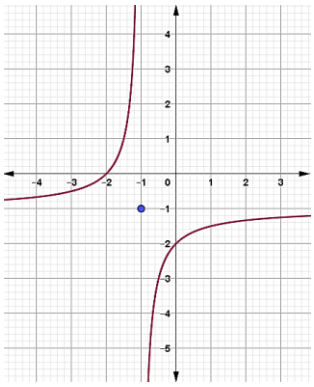
Example (4)

Choose the correct answer:

The opposite figure represents the curve of the function $f: f(x) = \ldots\ldots\ldots$

(a) $\frac{1}{x+1} - 1$ (b) $\frac{-1}{x+1} - 1$

(c) $\frac{-1}{x-1} + 1$ (d) $\frac{1}{x-1} + 1$



Solution

(b) The opposite figure represents the curve of the function $f: f(x) = \frac{-1}{x+1} - 1$

Exercise (3-6)

1- Write the symmetric point of the graph of each of the the following functions:

a) $f(x) = \frac{1}{x-2} + 3$

b) $f(x) = \frac{1}{x-1} + 4$

c) $f(x) = \frac{1}{x} + 2$

d) $f(x) = \frac{1}{x} - 5$

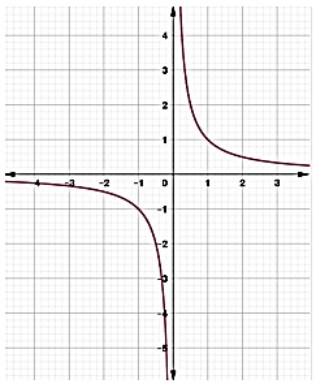
e) $f(x) = \frac{1}{x+4}$

f) $f(x) = -\frac{1}{x-3} + 2$

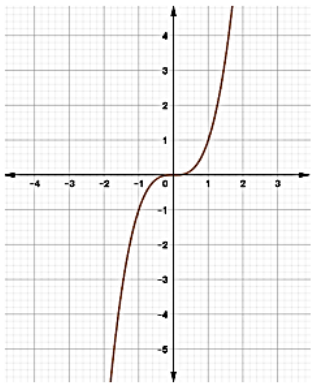
g) $f(x) = \frac{1}{x+2} - 1$

h) $f(x) = 2 - \frac{1}{x-5}$

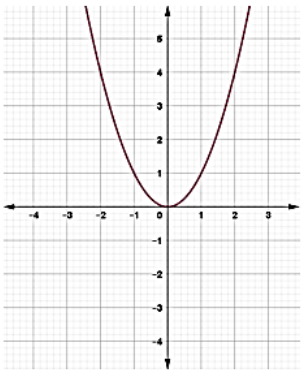
2- Match each function to its graph:



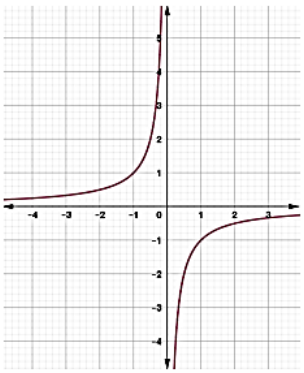
(a)



(b)



(c)



(d)

$f(x) = x^2$	$f(x) = -x^2$	$f(x) = x^3$	$f(x) = -x^3$	$f(x) = \frac{1}{x}$	$f(x) = -\frac{1}{x}$

3- Write the symmetric point of the graph of each of the the following functions:

a) $f(x) = \frac{1}{x} + 1$

b) $f(x) = \frac{1}{x-1}$

c) $f(x) = \frac{-1}{x}$

d) $f(x) = \frac{-1}{x-1}$

e) $f(x) = \frac{1}{x-2} + 1$

Exponential Function

The function f such that $f(x) = a^x, a > 0, a \neq 1, x \in R$ is called exponential function.

a is called the base, x is called the power.

Example (1)

Determine which of the following an exponential function is.

a) $f(x) = x^3$

b) $f(x) = 2^x$

c) $f(x) = \frac{3}{x-1}$

d) $f(x) = (-2)^x$

e) $f(x) = (\frac{1}{2})^{x-1}$

f) $f(x) = (x+3)^2$

Solution

a) $f(x) = x^3$ is not exponential

b) $f(x) = 2^x$ exponential

c) $f(x) = \frac{3}{x-1}$ is not exponential

d) $f(x) = (-2)^x$ is not exponential, the base is negative.

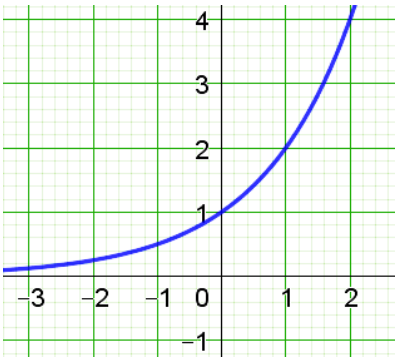
e) $f(x) = (\frac{1}{2})^{x-1}$ is an exponential

f) $f(x) = (x+3)^2$ is not exponential

The graph of the exponential function $f: f(x) = a^x, a > 0, a \neq 1, x \in R$

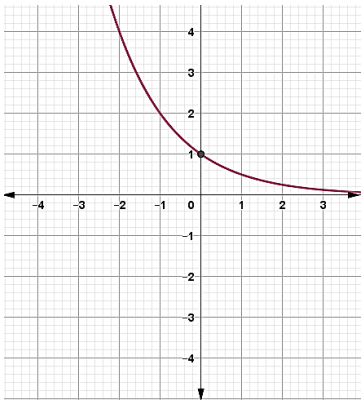
$f: f(x) = 2^x$, $x \in [-3, 3]$

x	-3	-2	-1	0	1	2	3
(x)	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8



$f: f(x) = (\frac{1}{2})^x$, $x \in [-3, 3]$

x	-2	-2	-1	0	1	2	3
(x)	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$



Notes

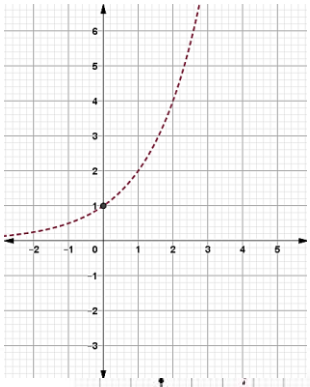
1) The domain of the exponential function $f{:}f(x) = a^x, a > 0, a \neq 1, x \in R$ is R and its range is]0,∞ [

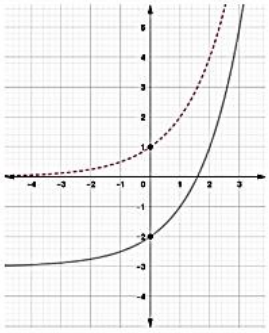
2) If a >1 then the function is increasing on its domain (growth)

3) If 0 < a < 1 then the function is decreasing on its domain (decay)

4) The curve of $f(x) = a^x$ passes through the point $(0, 1)$ for all $a > 0, a \neq 1$

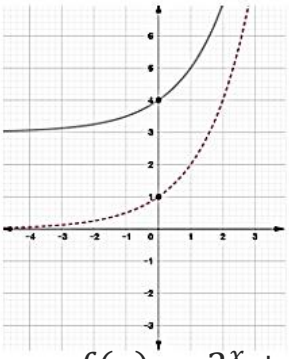
The opposite figure represents the graph of the function $f(x) = 2^x$, each of the following functions has a similar graph , but after certain translation





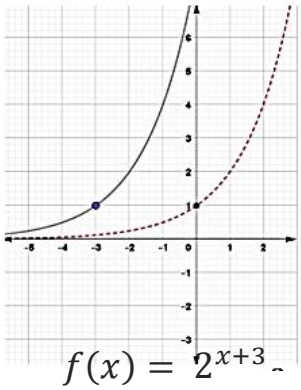
$f(x) = 2^x - 3$

3 units down



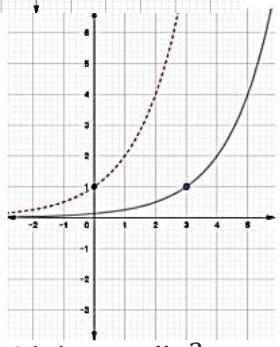
$f(x) = 2^x + 3$

3 units up



$f(x) = 2^{x+3}$

3 units to the left



$f(x) = 2^{x-3}$

3 units to the right

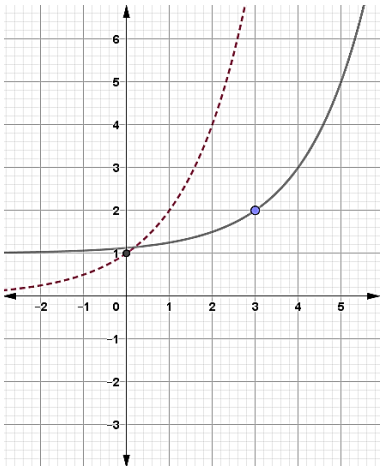
Example (1)

Choose the correct answer:

The opposite figure represents the curve of the function $f{:}\ f(x) = \ldots\ldots\ldots$

(a) $2^{x-1} + 3$ (b) $2^{x+1} + 3$

(c) $2^{x-3} + 1$ (d) $2^{x+3} + 1$



Solution

(c) $f(x) = 2^{x-3} + 1$

Exercise (3-7)

1- Complete each of the following:

a) The function f such that $f(x) = a^x$ is an exponential function if a ..... , x .....

b) The base of the exponential function f where $f(x) = 3^{x-1}$ is .....

c) The function f where K$f(x) = (-2)^{x-1}$ is not exponential because .....

d) The point of intersection of the curve of the function $f(x) = a^x$ with the y-axis is .....

2- Choose the correct answer:

i) The exponential function of base a is increasing if .....

(a) a > 0 (b) a > 1 (c) 0 < a < 1 (d) a = 1

ii) The exponential function of base a is decreasing if .....

(a) a > 0 (b) a < 0 (c) 0 <a < 1 (d) -1 <a < 0

iii) The exponential function $f(x) = a^x$, a > 1 its curve approaches .....

(a) positive direction of the x-axis . (b) negative direction of the x-axis

(c) positive direction of the y-axis (d) negative direction of the y-axis

3- Choose the correct answer:

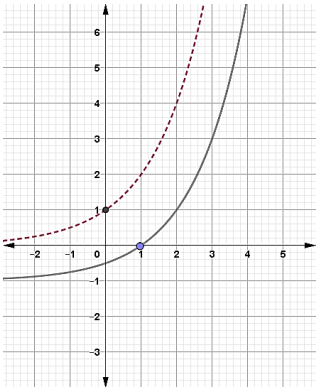
The opposite figure represents the curve of the function f: $f(x) =$ .....

(a) $2^{x-1} + 1$

(b) $2^{x+1} + 1$

(c) $2^{x-1} - 1$

(d) $2^{x+1} - 1$



4- Choose the correct answer:

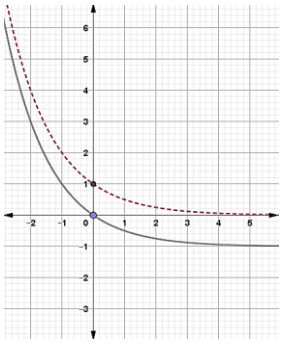
The opposite figure represents the curve of the function f: $f(x) =$ .....

(a) $2^x - 1$

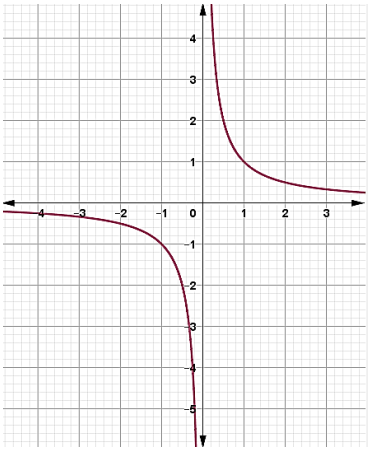
(b) $2^x + 1$

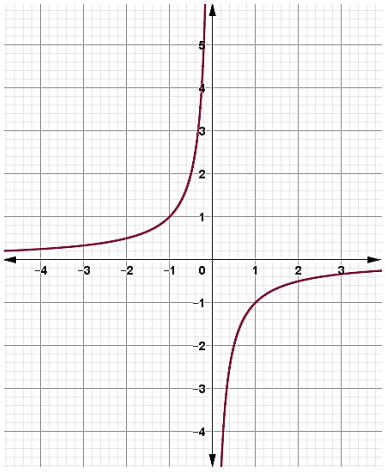
(c) $(\frac{1}{2})^x - 1$

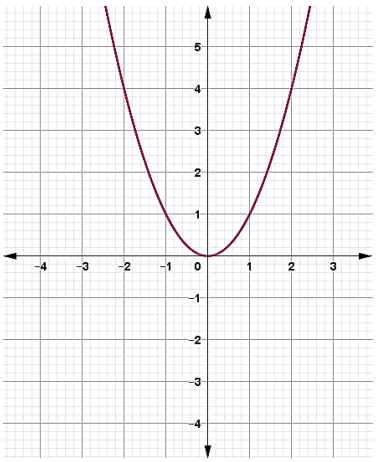
(d) $(\frac{1}{2})^x + 1$



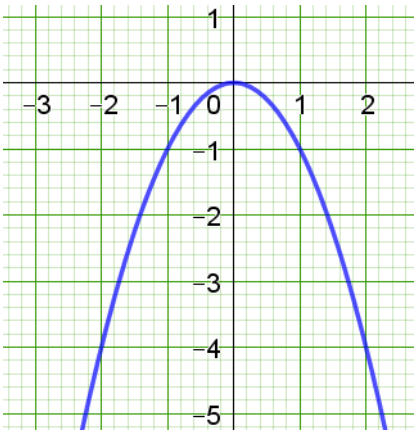
1-Write the rule of the function that shown in each graph:

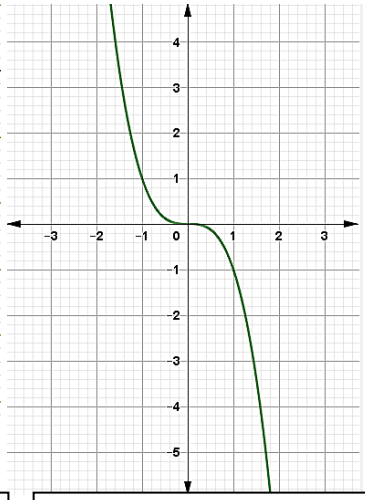


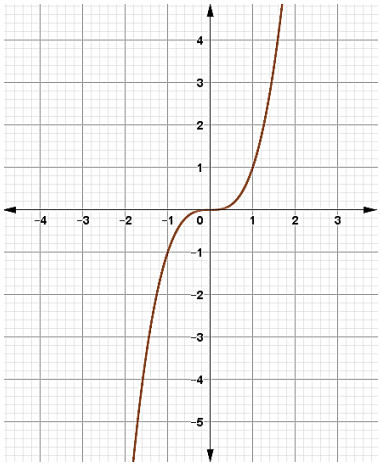




$a)\ f(x) =$	$b)\ f(x) =$	$c)\ f(x) =$ .....







$d)\ f(x) =$	$e)\ f(x) =$	$f)\ f(x) =$ .....

2-Write the symmetric point of each of the following functions:

a) $f(x) = x^2 + 1$
b) $f(x) = x^3 - 2$
c) $f(x) = \frac{1}{x+1}$
d) $f(x) = (x-1)^2 + 3$
e) $f(x) = (x+1)^2 - 1$

Unit 4	Lesson 1	Introduction to Limits

Types of quantities:

Defined quantities: real numbers as 4 , -2 , $\sqrt{3}$, 0 , $\frac{2}{3}$, ...

Undefined quantities: $\frac{\text{a real number} \neq 0}{\text{Zero}}$

Unspecified quantities: as $\frac{0}{0}$, $\frac{\infty}{\infty}$, $\infty - \infty$, ...

(Ask your teacher about the **7 unspecified** quantities)

Positive infinity (∞):

∞ indicates unbounded quantity larger than any real number that can be imagined

Negative infinity ($-\infty$):

$-\infty$ indicates unbounded quantity smaller than any real number that can be imagined

Notice that ∞ and $-\infty$ are not real numbers, $\mathrm{R} =]-\infty, \infty[$

Example (1)

Find (if possible) the result of each of the following operations:

a) $4 + \infty$ b) $3 - \infty$ c) $0 \div 3$ d) $-5 \div 0$

e) $\infty + \infty$ f) $0 \div 0$ g) $5 \times \infty$ h) $-6 \times -\infty$

Solution

a) $4 + \infty = \infty$ b) $3 - \infty = -\infty$ c) $0 \div 3 = 0$ d) $-5 \div 0$ **undefiend**

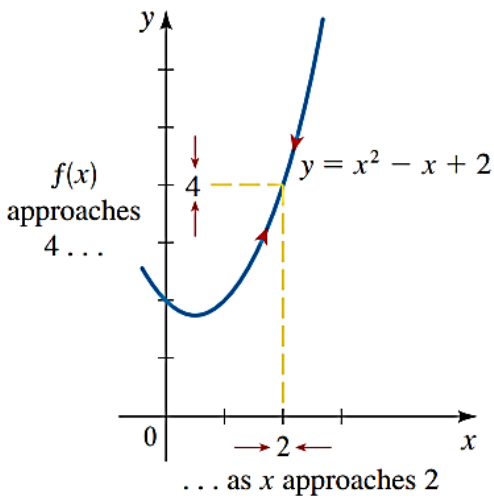
e) $\infty + \infty = \infty$ f) $0 \div 0$ **unspecified** g) $5 \times \infty = \infty$ h) $-6 \times -\infty = \infty$

Limits: The limit studies how function values (outputs) change as input values change.

Suppose that we have a function $\boldsymbol{f}$: $\boldsymbol{f}(x) = x^2 - x + 2$,

x	$f(x)$
1.0	2.000000
1.5	2.750000
1.8	3.440000
1.9	3.710000
1.95	3.852500
1.99	3.970100
1.995	3.985025
1.999	3.997001

x	$f(x)$
3.0	8.000000
2.5	5.750000
2.2	4.640000
2.1	4.310000
2.05	4.152500
2.01	4.030100
2.005	4.015025
2.001	4.003001



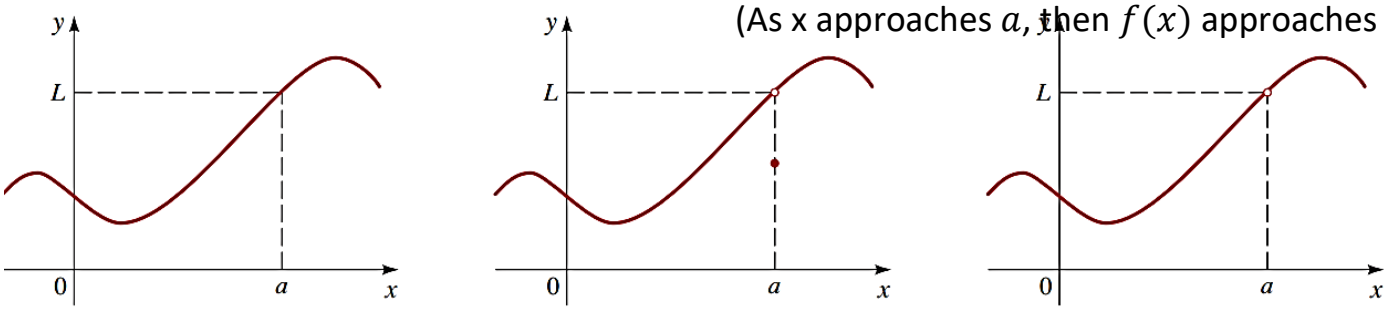
In the table and graph above, we see that as the input values approach 2 from the left (that is, are less than 2), the output values approach 4, and as the input values approach 2 from the right (that is, are greater than 2), the output values also approach 4.

Thus, we say: limit of $\boldsymbol{f}\,(x) = \; x^2 - x \; + 2$ as x approaches 2 is 4.

and it is written as $\lim\limits_{x\to 2}(\,x^2 - x \; + 2) = 4$

In general, the figure below shows that $\lim\limits_{x\to a} f(x) = L$

(As x approaches a, then $f(x)$ approaches L)



(a) (b) (c)

Example (1)

Evaluate the following limits:

a) $\lim\limits_{x\to 5}(\boldsymbol{2x^2 - 3x + 4})$

b) $\lim\limits_{x\to -2}\left(\frac{\boldsymbol{x^2+2x^2-1}}{\boldsymbol{5-3x}}\right)$

Solution

a) $$\begin{aligned}\lim_{x\to 5}(2x^2 - 3x + 4) &= \lim_{x\to 5}(2x^2) - \lim_{x\to 5}(3x) + \lim_{x\to 5} 4\\ &= 2\lim_{x\to 5} x^2 - 3\lim_{x\to 5} x + \lim_{x\to 5} 4\\ &= 2(5^2) - 3(5) + 4\\ &= 39\end{aligned}$$

b) $$\begin{aligned}\lim_{x\to -2}\frac{x^3 + 2x^2 - 1}{5 - 3x} &= \frac{\lim\limits_{x\to -2}(x^3 + 2x^2 - 1)}{\lim\limits_{x\to -2}(5 - 3x)}\\ &= \frac{\lim\limits_{x\to -2} x^3 + 2\lim\limits_{x\to -2} x^2 - \lim\limits_{x\to -2} 1}{\lim\limits_{x\to -2} 5 - 3\lim\limits_{x\to -2} x}\\ &= \frac{(-2)^3 + 2(-2)^2 - 1}{5 - 3(-2)} \quad = -\frac{1}{11}\end{aligned}$$

Exercise (4-1)

1) Find (if possible) the result of each of the following operations:

a) $1 - \infty$

b) $3 + \infty$

c) $0 \div (-2)$

d) $9 \div \infty$

e) $\infty - \infty$

f) $\infty \div 3$

g) $0 \times \infty$

h) $-3 \times \infty$

II-Find each of the following limits:

a) $\lim\limits_{x \to 1} (2x + 3)$

b) $\lim\limits_{x \to 2} (x^2 - 2x + 5)$

c) $\lim\limits_{x \to -1} (3)$

d) $\lim\limits_{x \to 3} (\sqrt{x + 1} + 2)$

e) $\lim\limits_{x \to 3} \frac{2x-1}{3x+1}$

To find $\lim\limits_{x\to a} f(x)$ you should find $f(a)$ ..... direct substitution

$= \text{k} \in \text{R}$ (defined quantity), then $\lim\limits_{x\to a} f(x)$ Exists and equals k

$= \frac{\text{k}}{0}$, $\text{k} \neq 0$ (undefined quantity), then $\lim\limits_{x\to a} f(x)$ Does not Exist

$= \frac{0}{0}$ (Unspecified quantity), then $\lim\limits_{x\to a} f(x) = ??????$

You Should use one of the following:

- Factoring
- Long division
- Multiplying by the conjugate
- Theorems

Example (1)

Find each of the following limits:

a) $\lim\limits_{x\to 2}(3x-1)$

b) $\lim\limits_{x\to 1}(7)$

c) $\lim\limits_{x\to 3}\frac{x-1}{x+1}$

d) $\lim\limits_{x\to 2}\frac{x^2-1}{x+1}$

e) $\lim\limits_{x\to 1}\frac{x^2-1}{x^2+x-2}$

f) $\lim\limits_{x\to 5}\frac{x^2-2x-15}{x^2-5x}$

Solution

a) $\lim\limits_{x\to 2}(3x-1)$, $f(2) = 2(3) - 1 = 6 - 1 = 5 \therefore \lim\limits_{x\to 2}(3x-1) = 5$

b) $\lim\limits_{x\to 1}(7)$ $f(1) = 7 \therefore \lim\limits_{x\to 1}(7) = 7$

c) $\lim\limits_{x\to 3}(\frac{x-1}{x+1})$,$f(3) = \frac{3-1}{3+1} = \frac{2}{4} = \frac{1}{2} \therefore \lim\limits_{x\to 3}(\frac{x-1}{x+1}) = \frac{1}{2}$

d) $\lim\limits_{x\to 2}\frac{x^2-1}{x+1}$ $f(2) = \frac{4-1}{2+1} = \frac{3}{3} = 1 \therefore \lim\limits_{x\to 3}(\frac{x-1}{x+1}) = 1$

e) $\lim\limits_{x\to 1}\frac{x^2-1}{x^2+x-2}$

$f(1) = \frac{1-1}{1+1-2} = \frac{0}{0}$ $(unspesified) \therefore \lim\limits_{x\to 1}\frac{x^2-1}{x^2+x-2} = \lim\limits_{x\to 1}\frac{(x-1)(x+1)}{(x+2)(x-1)} = \lim\limits_{x\to 1}\frac{(x-1)(x+1)}{(x+2)(x-1)} = \lim\limits_{x\to 1}\frac{(x+1)}{(x+2)} = \frac{2}{3}$

f) $\lim\limits_{x\to 5}\frac{x^2-2x-15}{x^2-5x}$

$f(1) = \frac{25-10-15}{25-25} = \frac{0}{0}$ $(unspesified) \therefore \lim\limits_{x\to 5}\frac{x^2-2x-15}{x^2-5x} = \lim\limits_{x\to 5}\frac{(x-5)(x+3)}{x(x-5)} = \lim\limits_{x\to 5}\frac{(x+3)}{x} = \frac{8}{5}$

Example (2)

Find each of the following limits:

a) $\lim\limits_{x\to 2}\frac{x^2+3x-10}{x^2-4}$

b) $\lim\limits_{x\to -3}\frac{x^2+x-6}{x^2+3x}$

Solution

a) $\lim\limits_{x\to 2}\frac{x^2+3x-10}{x^2-4}$

$f(2)=\frac{2^2+3(2)-10}{2^2-4}=\frac{0}{0}$ $(unspesified)$ $\therefore$ $\lim\limits_{x\to 2}\frac{x^2+3x-10}{x^2-4}=\lim\limits_{x\to 2}\frac{(x-2)(x+5)}{(x+2)(x-2)}=\lim\limits_{x\to 2}\frac{(x+5)}{(x+2)}=\frac{7}{4}$

b) $\lim\limits_{x\to -3}\frac{x^2+x-6}{x^2+3x}$

$f(3)=\frac{9-3-6}{9-9}=\frac{0}{0}$ $(unspesified)$ $\therefore$ $\lim\limits_{x\to -3}\frac{x^2+x-6}{x^2+3x}=\lim\limits_{x\to -3}\frac{(x-2)(x+3)}{x(x+3)}=\lim\limits_{x\to -3}\frac{(x-2)}{x}=\frac{-5}{-3}=\frac{5}{3}$

Example (3)

Find each of the following limits:

a) $\lim\limits_{x\to 3}\frac{\sqrt{x+1}-2}{x-3}$

b) $\lim\limits_{x\to 5}\frac{x-5}{\sqrt{x-1}-2}$

Solution

a) $\lim\limits_{x\to 3}\frac{\sqrt{x+1}-2}{x-3}=\lim\limits_{x\to 3}\frac{\sqrt{x+1}-2}{x-3}\times\frac{\sqrt{x+1}+2}{\sqrt{x+1}+2}=\lim\limits_{x\to 3}\frac{(x+1)-4}{(x-3)(\sqrt{x+1}+2)}$

$=\lim\limits_{x\to 3}\frac{(x-3)}{(x-3)(\sqrt{x+1}+2)}=\lim\limits_{x\to 3}\frac{1}{\sqrt{x+1}+2}=\frac{1}{4}$

b) $\lim\limits_{x\to 5}\frac{x-5}{\sqrt{x-1}-2}=\lim\limits_{x\to 5}\frac{x-5}{\sqrt{x-1}-2}\times\frac{\sqrt{x-1}+2}{\sqrt{x-1}+2}=\lim\limits_{x\to 5}\frac{(x-5)(\sqrt{x-1}+2)}{(\sqrt{x-1}-2)(\sqrt{x-1}+2)}=\lim\limits_{x\to 5}\frac{(x-5)(\sqrt{x-1}+2)}{(x-1)-4}=$

$\lim\limits_{x\to 5}\frac{(x-5)(\sqrt{x-1}+2)}{x-5}\lim\limits_{x\to 5}(\sqrt{x-1}+2)=4$

Exercise (4-2)

I-Choose the correct answer

1) $\lim_{x\to -1}\left(\frac{x+1}{x-3}\right)$ = ...

(a) 0 (b) does not exist (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

2) $\lim_{x\to 2}\left(\frac{x+2}{x-2}\right)$ = ...

(a) 0 (b) does not exist (c) 1 (d) -1

3) $\lim_{x\to -2}\left(\frac{x^2}{x+2}+\frac{2x}{x+2}\right)$ = ...

(a) 0 (b) $\frac{1}{2}$ (c) -1 (d) -2

4) $\lim_{x\to 1}\left(\frac{x^2+x-2}{x^2-3x+2}\right)$= .....

(a) $-\frac{2}{3}$ (b) $\frac{2}{3}$ (c) -1 (d) -3

5) $\lim_{x\to 9}\left(\frac{x^2-81}{\sqrt{x}-3}\right)$= .....

(a) 9 (b) 10 (c) 18 (d) 108

6) $\lim_{x\to 0}\left(\frac{x}{\sqrt{x+4}-2}\right)$ = ...

a. (a) 0 (b) 2 (c) 4 (d) does not exist

7) $\lim_{x\to 3}\left(\frac{x^2-3x}{x^2-2x-3}\right)$= .....

(a) 0 (b) 5 (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

8) $\lim_{x\to 3}\left(\frac{x-3}{x^2-8x+15}\right)=\frac{1}{a}$, then a = ...

(a) - 2 (b) 2 (c) $\frac{1}{2}$ (d) $\frac{-1}{2}$

II-Find each of the following limits:

$a)\ \lim_{x\to 2}\frac{x^2-5x+6}{x^2-4}$

$b)\ \lim_{x\to 3}\frac{x^2-5x+6}{x^2-4x+3}$

$c)\ \lim_{x\to 1}\frac{x^2-5x+4}{x^3-1}$

$d)\ \lim_{x\to 3}\frac{x^2-3x}{x^2-9}$

e) $\lim_{x \to -4} \frac{x^2 + 5x + 4}{x^2 + 3x - 4}$

f) $\lim_{x \to 2} \frac{x^2 + x - 6}{x - 2}$

g) $\lim_{x \to 1} \frac{x^3 - 1}{x^2 - 1}$

h) $\lim_{x \to 2} \frac{x^2 - x + 6}{x + 2}$

i) $\lim_{h \to 0} \frac{\sqrt{1 + h} - 1}{h}$

J) $\lim_{t \to -3} \frac{t^2 - 9}{2t^2 + 7t + 3}$

k) $\lim_{x \to 2} \frac{x^4 - 16}{x - 2}$

l) $\lim_{h \to 0} \frac{(2 + h)^3 - 8}{h}$

m) $\lim_{x \to 1} \frac{x^2 - 1}{\sqrt{x} - 1}$

n) $\lim_{x \to 7} \frac{\sqrt{x + 2} - 3}{x - 7}$

o) $\lim_{x \to -1} \frac{x^2 - x - 2}{x^3 - x}$

Theorem $\lim\limits_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}$

Example (1)

Find each of the following limits:

a) $\lim\limits_{x \to 2} \frac{x^5 - 32}{x - 2}$ b) $\lim\limits_{x \to -3} \frac{x^4 - 81}{x + 3}$

Solution

a) $\lim\limits_{x \to 2} \frac{x^5 - 32}{x - 2}$ $f(2) = \frac{2^5 - 32}{x - 2} = \frac{0}{0}$ unspecified

$$\lim_{x \to 2} \frac{x^5 - 32}{x - 2} = \lim_{x \to 2} \frac{x^5 - 2^5}{x - 2} = 5(2)^{5-1} = 80$$

$b)$ $\lim\limits_{x \to -3} \frac{x^4 - 81}{x + 3}$ $f(-3) = \frac{(-3)^4 - 81}{3 - 3} = \frac{0}{0}$ unspecified

$$\lim_{x \to -3} \frac{x^4 - 81}{x + 3} = \lim_{x \to -3} \frac{x^4 - (-3)^4}{x - (-3)} = 4(-3)^{4-1} = -108$$

Corollary 1: $\lim\limits_{x \to a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}$

Example (2)

Find each of the following limits:

a) $\lim\limits_{x \to 2} \frac{x^7 - 128}{x^3 - 8}$ b) $\lim\limits_{x \to \frac{1}{2}} \frac{128x^7 - 1}{8x^3 - 1}$

Solution

$\lim\limits_{x \to 2} \frac{x^7 - 128}{x^3 - 8}$ $f(2) = \frac{2^7 - 128}{2^3 - 8} = \frac{0}{0}$ unspecified

$$\lim_{x \to 2} \frac{x^7 - 128}{x^3 - 8} = \lim_{x \to 2} \frac{x^7 - 2^7}{x^3 - 2^3} = \frac{7}{3}(2)^{7-3} = \frac{7}{3}(16) = \frac{112}{3}$$

$b)$ $\lim\limits_{x \to \frac{1}{2}} \frac{128x^7 - 1}{8x^3 - 1}$ $f\left(\frac{1}{2}\right) = \frac{128(\frac{1}{2})^7 - 1}{8(\frac{1}{2})^3 - 1} = \frac{0}{0}$ *unspecified*

$$\lim_{x \to \frac{1}{2}} \frac{128x^7 - 1}{8x^3 - 1} = \lim_{x \to \frac{1}{2}} \frac{128[x^7 - \left(\frac{1}{2}\right)^7]}{8[x^3 - \left(\frac{1}{2}\right)^3]} = 16 \times \frac{7}{3}(\frac{1}{2})^{7-3} = \frac{7}{3}$$

Another solution:

$$\lim_{x \to \frac{1}{2}} \frac{128x^7-1}{8x^3-1} = \lim_{2x \to 1} \frac{(2x)^7-1^7}{(2x)^3-1^3} = \frac{7}{3}(1)^{7-3} = \frac{7}{3}$$

Corollary 2: $\lim_{x \to 0} \frac{(x+a)^n-a^n}{x} = na^{n-1}$

Example (3)

Find each of the following limits:

a) $\lim_{x \to 0} \frac{(x+3)^5-243}{x}$

b) $\lim_{x \to 0} \frac{(x+1)^8-1}{5x}$

Solution

a) $\lim_{x \to 0} \frac{(x+3)^5-243}{x}$ $\qquad f(0) = \frac{3^5-243}{0} = \frac{0}{0}$ unspecified

$$\lim_{x \to 0} \frac{(x+3)^5-243}{x} = \lim_{x \to 0} \frac{(x+3)^5-(3)^5}{x} = 5(3)^4 = 405$$

b) $\lim_{x \to 0} \frac{(x+1)^8-1}{5x}$ $\qquad f(0) = \frac{1^8-1}{0} = \frac{0}{0}$ unspecified

$$\lim_{x \to 0} \frac{(x+1)^8-1}{5x} = \frac{1}{5} \lim_{x \to 0} \frac{(x+1)^8-(1)^8}{x} = 8(1)^7 = \frac{8}{5}$$

Exercise (4-3)

Find each of the following limits:

a) $\lim\limits_{x \to 2} \dfrac{x^5 - 32}{x - 2}$

b) $\lim\limits_{x \to -3} \dfrac{x^4 - 81}{x^5 + 243}$

c) $\lim\limits_{x \to 2} \dfrac{x^{-9} - \frac{1}{512}}{x - 2}$

d) $\lim\limits_{x \to -3} \dfrac{\sqrt{x + 7} - 2}{x + 3}$

e) $\lim\limits_{x \to 2} \dfrac{x^7 - 128}{x^5 - 32}$

f) $\lim\limits_{x \to 1} \dfrac{\sqrt[7]{x} - 1}{x - 1}$

g) $\lim\limits_{x \to 2} \dfrac{(x - 3)^6 - 1}{x - 2}$

h) $\lim\limits_{x \to 81} \dfrac{\sqrt[4]{x} - 3}{x - 81}$

i) $\lim\limits_{x \to 3} \dfrac{(x + 1)^4 - 256}{x - 3}$

j) $\lim\limits_{x \to 7} \dfrac{(x - 2)^4 - 625}{x - 7}$

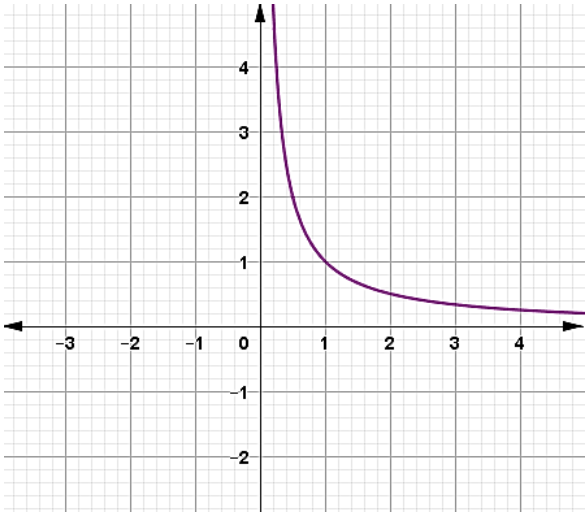
k) $\lim\limits_{x \to 3} \dfrac{(2x - 1)^5 - 3125}{x - 3}$

l) $\lim\limits_{x \to 3} \dfrac{x^6 - 729}{x^5 - 243}$

Limit of a function at infinity

The opposite figure represents the graph of the function ***f*** where: $f(x) = \frac{1}{x}, x >$

we notice from the graph when x approaches ∞, then $f(x) = \frac{1}{x}$, approaches zero.



We deduce that:

$\lim\limits_{x \to \infty} (\frac{1}{x}) = 0$

$\lim\limits_{x \to \infty} (\frac{a}{x}) = 0$, a is constant.

$\lim\limits_{x \to \infty} (a) = a$, a is constant.

$\lim\limits_{x \to \infty} (x^n) = \infty$, n is positive integer.

Example (1)

Find each of the following limits:

a) $\lim\limits_{x \to \infty} (\frac{1}{x} + 3)$

b) $\lim\limits_{x \to \infty} (4 - \frac{3}{x^2})$

Solution

a) $\lim\limits_{x \to \infty} (\frac{1}{x} + 3) = \lim\limits_{x \to \infty} \frac{1}{x} + \lim\limits_{x \to \infty} 3$

$= 0 + 3 = 3$

$\therefore \lim\limits_{x \to \infty} (\frac{1}{x} + 3) = 3$

b) $\lim\limits_{x \to \infty} (4 - \frac{3}{x^2}) = \lim\limits_{x \to \infty} 4 - \lim\limits_{x \to \infty} \frac{3}{x^2}$

$= 4 - 3 \lim\limits_{x \to \infty} \frac{1}{x^2} = 4 - 3 \times 0 = 4$

$\therefore \lim\limits_{x \to \infty} (4 - \frac{3}{x^2}) = 4$

Example (2)

Find each of the following limits:

a) $\lim\limits_{x \to \infty} \left(\frac{2x-3}{3x^2+1}\right)$ b) $\lim\limits_{x \to \infty} \left(\frac{2x^2-3}{3x^2+1}\right)$ c) $\lim\limits_{x \to \infty} \left(\frac{2x^3-3}{3x^2+1}\right)$

a) $$\lim_{x \to \infty} \frac{2x - 3}{3x^2 + 1} = \frac{\lim\limits_{x \to \infty} \left(\frac{2}{x} - \frac{3}{x^2}\right)}{\lim\limits_{x \to \infty} \left(3 + \frac{1}{x^2}\right)} = \frac{0 - 0}{3 + 0} = 0$$

b) $$\lim_{x \to \infty} \frac{2x^2 - 3}{3x^2 + 1} = \frac{\lim\limits_{x \to \infty} \left(2 - \frac{3}{x^2}\right)}{\lim\limits_{x \to \infty} \left(3 + \frac{1}{x^2}\right)}$$

$$= \frac{2 - 0}{3 + 0} = \frac{2}{3}$$

c) $$\lim_{x \to \infty} \frac{2x^3 - 3}{3x^2 + 1} = \frac{\lim\limits_{x \to \infty} \left(2x - \frac{3}{x^2}\right)}{\lim\limits_{x \to \infty} \left(3 + \frac{1}{x^2}\right)}$$

$$= \frac{\infty - 0}{3 + 0} = \infty$$

Example (3)

Find each of the following limits: $\lim\limits_{x \to \infty} \left(\frac{3x^2-x-2}{5x^2+4x+1}\right)$

Solution Divide each of the denominator and numerator by x^2

$$\lim_{x \to \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1} = \lim_{x \to \infty} \frac{3 - \frac{1}{x} - \frac{2}{x^2}}{5 + \frac{4}{x} + \frac{1}{x^2}}$$

$$= \frac{\lim\limits_{x \to \infty} \left(3 - \frac{1}{x} - \frac{2}{x^2}\right)}{\lim\limits_{x \to \infty} \left(5 + \frac{4}{x} + \frac{1}{x^2}\right)}$$

$$= \frac{3 - 0 - 0}{5 + 0 + 0} = \frac{3}{5}$$

Exercise (4-4)

Find each of the following limits:

1) $\lim\limits_{x \to \infty} (3x^{-5} + 4x^{-2} + 5)$

2) $\lim\limits_{x \to \infty} 3$

3) $\lim\limits_{x \to \infty} \left(\frac{3}{x^2} - 2\right)$

4) $\lim\limits_{x \to \infty} 2x^{-5}$

5) $\lim\limits_{x \to \infty} \frac{2x^2 + 5x + 1}{3x^2 + 8x - 3}$

6) $\lim\limits_{x \to \infty} \frac{5 - 6x - 3x^2}{2x^2 + x + 4}$

7) $\lim\limits_{x \to \infty} \frac{7x^2 + 1}{4x^3 - 8x + 1}$

8) $\lim\limits_{x \to \infty} \frac{5x^7 + 2x - 1}{6x^4 + 13}$

9) $\lim\limits_{x \to \infty} \frac{x^3 - 4x + 5}{(2x - 1)^3}$

10) $\lim\limits_{x \to \infty} \frac{4 - 3x^3}{\sqrt{x^6 + 9}}$

11) $\lim\limits_{x \to \infty} \frac{\sqrt[3]{8x^2 + 1}}{\sqrt[6]{x^4 - 3}}$

12) $\lim\limits_{x \to \infty} \left(\frac{x}{2x + 1} + \frac{3x^2}{(x - 3)^2}\right)$

13) $\lim\limits_{x \to \infty} \frac{\sqrt{4x^2 + 7} + 3x}{2x + 9}$

14) $\lim\limits_{x \to \infty} \left(\sqrt{x^2 + x - 1} - \sqrt{x^2 - x + 1}\right)$

15) $\lim\limits_{x \to \infty} \frac{5x^{-3} + 4x^{-2} - 3}{7x^{-3} - 2x^{-2} + 8}$

16) $\lim\limits_{x \to \infty} \frac{(2x - 1)^5 (x^2 + 3)^6}{(x + 1)^7 (x^2 - 5)^5}$

The right limit and the left limit

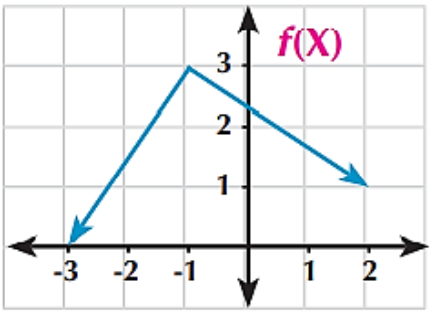
The limit of the function ***f*** when x tends to *a* equals L if and only if the right limit and the left limit when x tends to *a* are equal and each of them equals L where L ∈ R,

- The right limit of the function is $f(a^+) = \lim\limits_{x \to a^+} f(x)$
- The left limit of the function is $f(a^-) = \lim\limits_{x \to a^-} f(x)$
- $\lim\limits_{x \to a} f(x)$ exists and equals L if and only if $f(a^+) = f(a^-) = L$

Example (1)

From the opposite graph, complete:

a) $f(-1) =$ .....
b) $\lim\limits_{x \to -1^+} f(x) =$ .....
c) $\lim\limits_{x \to -1^-} f(x) =$ .....
d) $\lim\limits_{x \to -1} f(x) =$ .....



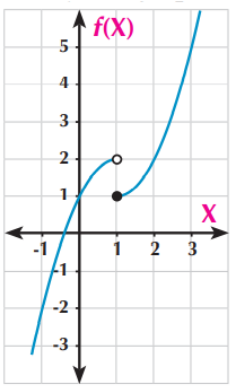
Solution a) $f(-1) = 3$ b) $\lim\limits_{x \to -1^+} f(x) = 3$

c) $\lim\limits_{x \to -1^-} f(x) = 3$ d) $\lim\limits_{x \to -1} f(x) = 3$

Example (2)

From the opposite graph, complete:

a) $f(1) =$ .....
b) $\lim\limits_{x \to 1^+} f(x) =$ .....
c) $\lim\limits_{x \to 1^-} f(x) =$ .....
d) $\lim\limits_{x \to 1} f(x) =$ .....

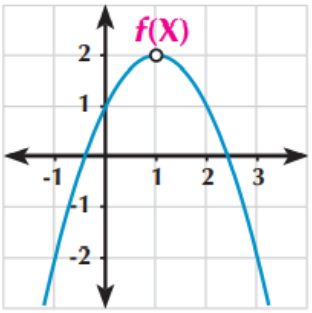


Solution a) $f(1) = 1$ b) $\lim\limits_{x \to 1^+} f(x) = 1$

c) $\lim\limits_{x \to 1^-} f(x) = 2$ d) $\lim\limits_{x \to 1} f(x)$ does not exist. $[\lim\limits_{x \to 1^+} f(x) \neq \lim\limits_{x \to 1^-} f(x)]$

Example (3)

From the opposite graph, complete:



a) $f(1) =$ .....

b) $\lim\limits_{x \to 1^+} f(x) =$ .....

c) $\lim\limits_{x \to 1^-} f(x) =$ .....

d) $\lim\limits_{x \to 1} f(x) =$ .....

Solution a) $f(1) =$ (undefined)

b) $\lim\limits_{x \to 1^+} f(x) = 2$

c) $\lim\limits_{x \to 1^-} f(x) = 2$

d) $\lim\limits_{x \to 1} f(x) = 2$,

$[\lim\limits_{x \to 1^+} f(x) = \lim\limits_{x \to 1^-} f(x) = 2]$

Example (4)

If ***f*** is a function such that $f(x) = \begin{cases} 8 - x & \text{at } x \geqslant 3 \\ x + 2 & \text{at } x < 3 \end{cases}$, then complete:

a) $f(3) =$ .....

b) $\lim\limits_{x \to 3^+} f(x) =$ .....

c) $\lim\limits_{x \to 3^-} f(x) =$ .....

d) $\lim\limits_{x \to 3} f(x) =$ .....

Solution

a) $f(3) = 8 - 3 = 5$

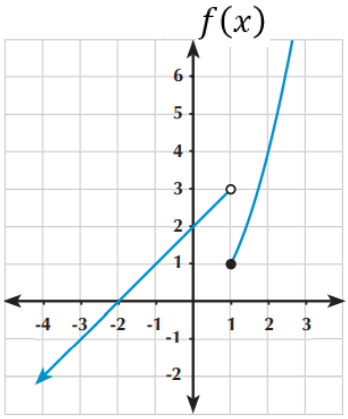
b) $\lim\limits_{x \to 3^+} f(x) = \lim\limits_{x \to 3} (8 - x) = 5$

c) $\lim\limits_{x \to 3^-} f(x) = \lim\limits_{x \to 3} (x + 2) = 5$

d) $\lim\limits_{x \to 3} f(x) = 5$ $[\lim\limits_{x \to 3^+} f(x) = \lim\limits_{x \to 3^-} f(x) = 5]$

Exercise (4-5)

1- From the opposite graph, complete:



a) $f(1) =$ .....

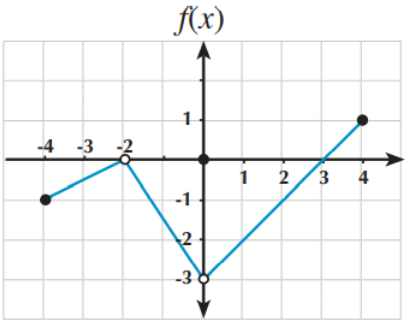
b) $\lim\limits_{x \to 1^+} f(x) =$ .....

c) $\lim\limits_{x \to 1^-} f(x) =$ .....

d) $\lim\limits_{x \to 1} f(x) =$ .....

e) $\lim\limits_{x \to -1} f(x) =$ .....

2- From the opposite graph, complete:



a) $\lim\limits_{x \to -4} f(x) =$ .....

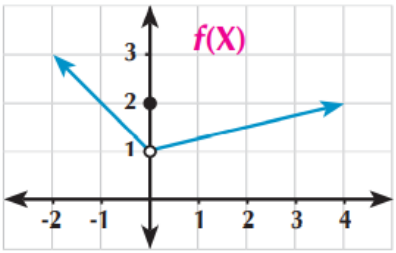
b) $\lim\limits_{x \to -2} f(x) =$ .....

c) $\lim\limits_{x \to 0} f(x) =$ .....

d) $\lim\limits_{x \to 1} f(x) =$ .....

e) $\lim\limits_{x \to 4} f(x) =$ .....

3- From the opposite graph, complete:



a) $\lim\limits_{x \to 0^-} f(x) =$ .....

b) $\lim\limits_{x \to 0^-} f(x) =$ .....

c) $f(0) =$ .....

d) $\lim\limits_{x \to -1} f(x) =$ .....

e) $f(-1) =$ .....

4- If $\lim\limits_{x \to 2} f(x) = 7$, where $f(x) = \begin{cases} x^2 + a & , x < 2 \\ 5x + b & , x > 2 \end{cases}$, then find the value of a , b

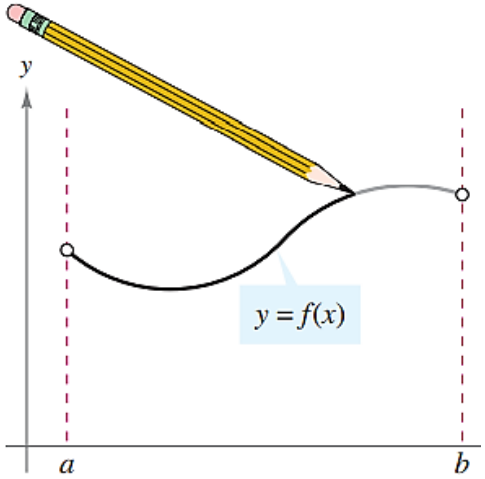
5- If $\lim\limits_{x \to -1} f(x)$ exists, where $f(x) = \begin{cases} x^2 + a & , x < -1 \\ x + 4 & , x > -1 \end{cases}$, then find:

i) the value of a

ii) $\lim\limits_{x \to -1} f(x)$

Continuity

Many functions have the property that their graphs can be traced with a pencil without lifting the pencil from the page. Such functions are called continuous. Other functions have points at which a break in the graph occurs but satisfy this property over intervals contained in their domains. They are continuous on these intervals and are said to have a discontinuity at a point where a break occurs.



Continuity of a function at a point:

The function ***f*** is continuous at $x = a$ if:

- ***f*** is defined at $x = a$ $f(a)$ is defined
- $\lim\limits_{x \to a} f(x)$ exists
- $\lim\limits_{x \to a} f(x) = f(a)$

A function ***f*** is discontinuous at a point a if it fails to be continuous at a.

Determining Continuity at a Point

1. Check to see if $f(a)$ is defined.
2. If $f(a)$ is undefined, then the function is not continuous at a.
3. If $f(a)$ is defined, Compute $\lim\limits_{x \to a} f(x)$. If $\lim\limits_{x \to a} f(x)$ does not exist, then the function is discontinuous at a and the problem is solved.
4. If $\lim\limits_{x \to a} f(x)$ exists, then compare $f(a)$ and $\lim\limits_{x \to a} f(x)$.
5. If $\lim\limits_{x \to a} f(x) \neq f(a)$, then the function is discontinuous at a.
6. If $\lim\limits_{x \to a} f(x) = f(a)$, then the function is continuous at a.

Continuity of a function on an interval:

The function ***f*** is continuous on the interval $]a, b[$ if it is continues at every point $x \in]a, b[$

Example (1)

Discuss the Continuity of the function f: $f(x)=\begin{cases} 2x+3, \text{ if } x\leq 2 \\ 2x-3, \text{ if } x>2 \end{cases}$ at $x=2$

Solution

$f(2)=2(2)+3=7$

$\lim\limits_{x\to 2^+} f(x)=\lim\limits_{x\to 2}(2x-3)=1$

$\lim\limits_{x\to 2^-} f(x)=\lim\limits_{x\to 2}(2x+3)=7$

$\lim\limits_{x\to 2^+} f(x)\neq \lim\limits_{x\to 2^-} f(x) \therefore \lim\limits_{x\to 2} f(x)$ does not exist $\therefore f(x)$ is discontinuous at $x=2$

Example (2)

Discuss the Continuity of the function f: $f(x)=\begin{cases} x^3-3, \text{ if } x\leq 2 \\ x^2+1, \text{ if } x>2 \end{cases}$ at $x=2$

Solution

$f(2)=2^3-3=5$

$\lim\limits_{x\to 2^+} f(x)=\lim\limits_{x\to 2}(x^2+1)=4+1=5$

$\lim\limits_{x\to 2^-} f(x)=\lim\limits_{x\to 2}(x^3-3)=8-3=5$

$\lim\limits_{x\to 2^+} f(x)=\lim\limits_{x\to 2^-} f(x)=5$

$\therefore \lim\limits_{x\to 2} f(x)=5$

$\therefore \lim\limits_{x\to 2} f(x)=f(2)=5$

$\therefore f(x)$ is continuous at $x=2$

Example (3)

Discuss the Continuity of the function f: $f(x) = \begin{cases} \dfrac{x^2 - 4}{x - 2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$ at $x = 2$

Solution

$f(2) = 4$

$$\lim_{x\to 2} f(x) = \lim_{x\to 2} \frac{x^2 - 4}{x - 2} = \lim_{x\to 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{x-2}} = \lim_{x\to 2}(x+2) = 4.$$

$$f(2) = 4 = \lim_{x\to 2} f(x)$$

$\therefore$ $\boldsymbol{f(x)}$ is continuous at $\boldsymbol{x = 2}$.

Example (4)

Find the relationship between a and b so that the function f: $\mathbf{f(x)} = \begin{cases} \mathbf{ax + 1, \text{ if } x \leq 3} \\ \mathbf{bx + 3, \text{ if } x > 3} \end{cases}$ is continuous at $x = 3$

Solution

$f(x)$is continuous at $x = 3$, then $\lim\limits_{x\to 3} f(x)$ exists

$$\lim_{x\to 3^+} f(x) = \lim_{x\to 3^-} f(x)$$

$$\lim_{x\to 3}(ax + 1) = \lim_{x\to 3}(bx + 3)$$

$$3a + 1 = 3b + 3$$

$$\therefore 3a = 3b + 2$$

Exercise (4-6)

1- Prove that the function $f(x) = 5x - 3$ is continuous at $x = 0$, and at $x = \mathbf{5}$.

2-Examine the continuity of the function $f(x) = 2x^2 - 1$ at $x = 3$.

3-Discuss the continuity of the function f: $f(x) = \begin{cases} x^2 + 2 , & x \leqslant 1 \\ 3x , & x > 1 \end{cases}$ at $x = 1$

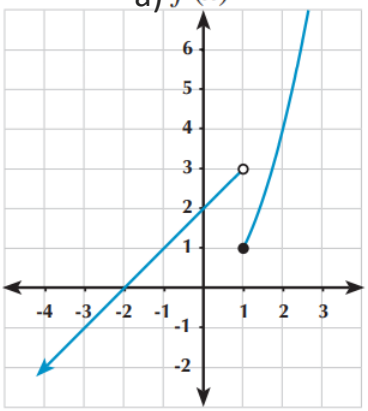
4-Discuss the continuity of the function f: $f(x) = \begin{cases} x^2 - 3x + 2 , & x \leqslant 3 \\ x^2 - 2x - 1 , & x > 3 \end{cases}$ at $x = 3$

5-Discuss the continuity of the function f: $f(x) = \begin{cases} x^2 - 1 , & x \leqslant 2 \\ 2x - 1 , & x > 2 \end{cases}$ at $x = 2$

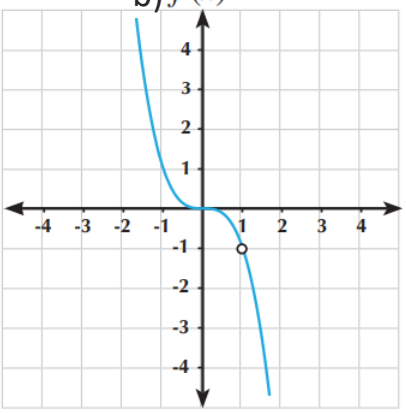
6-Discuss the continuity of the function f: $f(x) = \begin{cases} x + 4 , & x < -2 \\ 1 , & -2 \leqslant x < -1 \\ 2x + 3 , & x \geqslant -1 \end{cases}$ at $x = -2$, at $x = -1$

7-Discuss the continuity of the function f in each of the following figures.

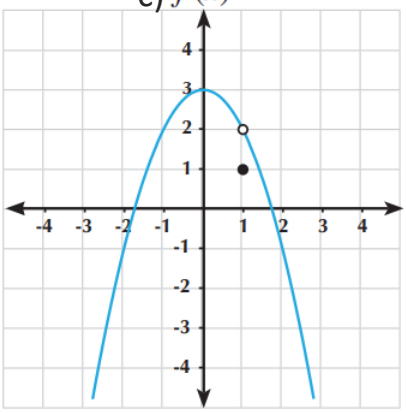
a) $f(x)$



b) $f(x)$



c) $f(x)$



Example (1)

Discuss the Continuity of the function f: $f(x) = \begin{cases} x - 2 & \text{when } x < 0 \\ x^2 & \text{when } 0 \leqslant x \leqslant 2 \\ 2x & \text{when } x > 2 \end{cases}$ on R.

Solution

i) at $x = 0$

$f(0) = 0^2 = 0$

$\lim\limits_{x \to 0^+} f(x) = \lim\limits_{x \to 0} (x^2) = 0^2 = 0$

$\lim\limits_{x \to 0^-} f(x) = \lim\limits_{x \to 0} (x - 2) = 0 - 2 = -2$

$\lim\limits_{x \to 0^+} f(x) \neq \lim\limits_{x \to 0^-} f(x)$ $\therefore f(x)$ is discontinuous at $x = 0$

ii) at $x = 2$

$f(2) = 2^2 = 4$

$\lim\limits_{x \to 2^+} f(x) = \lim\limits_{x \to 2} (2x) = 2(2) = 4$

$\lim\limits_{x \to 2^-} f(x) = \lim\limits_{x \to 2} (x^2) = 2^2 = 4$

$\lim\limits_{x \to 2^+} f(x) = \lim\limits_{x \to 2^-} f(x) = 4$

$\therefore \lim\limits_{x \to 2} f(x) = 4$ $\therefore \lim\limits_{x \to 2} f(x) = f(2) = 4$ $\therefore f(x)$ is continuous at $x = 2$

$when\ x < 0, f(x) = \mathrm{x} - 2$ is continuous

$when\ 0 < x < 2, f(x) = x^2$ is continuous

$when\ x > 2, f(x) = 2x$ is continuous

$\therefore f(x)$ is continuous on R – { 0 }

Exercise (4-7)

1) Discuss the continuity of the functions f: $f(x) = \begin{cases} 2x + 3 & \text{when } x \leqslant 4 \\ 7 + \frac{16}{x} & \text{when } x > 4 \end{cases}$ at $x = 4$

2) Find the values of k, m which make the function f continuous on **R** where

$$f(x) = \begin{cases} x^2 + 5 & \text{when } x \geqslant 2 \\ m(x + 1) + k & \text{when } -1 < x < 2 \\ 2x^3 + x + 7 & \text{when } x \leqslant -1 \end{cases}$$

3) Find the values of k which make the function f : $f(x) = \begin{cases} 7x - 2 & x \leqslant 1 \\ -k\,x^2 & x > 1 \end{cases}$ continuous on **R**

4) Find the values of k which make the function f : $f(x) = \begin{cases} k\,x^2 & x \leqslant 2 \\ 2x + k & x > 2 \end{cases}$ continuous on **R**

5) Find the values of k which make the function f : $f(x) = \begin{cases} 9 - x^2 & x \geqslant -3 \\ \frac{k}{x^2} & x < -3 \end{cases}$ continuous on **R**

6) Discuss the continuity of the function f : $f(x) = \begin{cases} x^2 + 1 , & x \geq 2 \\ x + 3 , & x < 2 \end{cases}$ on **R**

7) Discuss the continuity of the function f : $f(x) = \begin{cases} x^2 + 5x + 6 \\ 6 - x \end{cases}$ on **R**

I-Find each of the following limits:

1) $\lim\limits_{x \to 5} \dfrac{x - 5}{x^2 - 10x + 25}$

2) $\lim\limits_{x \to 1} \dfrac{x^5 - 1}{x^3 - 1}$

3) $\lim\limits_{x \to 0} \dfrac{\sqrt{25 + x} - 5}{\sqrt{1 + x} - 1}$

4) $\lim\limits_{x \to 1} \dfrac{x - 1}{x^2 + x - 2}$

5) $\lim\limits_{x \to \frac{3}{2}} \dfrac{2x^2 + 3x - 9}{2x - 3}$

6) $\lim\limits_{x \to 1} \dfrac{x^4 - \sqrt{x + 15}}{1 - x^2}$

7) $\lim\limits_{x \to 0} \dfrac{(4 + x)^3 - 64}{x}$

8) $\lim\limits_{x \to 1} \dfrac{x^8 - 1}{x^5 - x}$

9) $\lim\limits_{x \to \infty} \dfrac{3x^2 - 2}{x^2 + 1}$

10) $\lim\limits_{x \to \infty} \dfrac{3x^{-1} + 5}{7 + 9x^{-2}}$

11) $\lim\limits_{x \to \infty} \dfrac{1}{x} \sqrt{4x^2 - 3}$

II- If the function $f: f(x) = \begin{cases} \frac{x^2-1}{x-1} & , x \neq 1 \\ 2a & , x = 1 \end{cases}$ is continuous at $x = 1$,

then find the value of a

III- If the function $f: f(x) = \begin{cases} ax^2 - 4 & , x \neq 2 \\ 2a & , x = 2 \end{cases}$ is continuous at $x = 2$,

then find the value of a

IV- If the function $f: f(x) = \begin{cases} x^2 - 2x & , x \geq 1 \\ ax - 3 & , x < 1 \end{cases}$ is continuous at $x = 1$, then

find the value of a

V- If the function *f* where: ***f*(x) =** $\begin{cases} \mathbf{x^2 + ax - 1} & , x < 2 \\ \mathbf{5} & , x = 2 \\ a + bx & , x > 2 \end{cases}$ is continuous at x = 2 , find the values of the constants a and b

VI- If the function f(x) $= \begin{cases} \frac{\sqrt{x+3} - 2}{x-1} & , x \neq 1 \\ a & , x = 1 \end{cases}$ is continuous at x = 1, then find the value of a.